

Manual

Shop Floor / Order Data Management BDE-BDM 8.2

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1 Overview Shop Floor / Order Data Management

Purpose

The function package “shop floor / order data management” provides an extensive range functions for recording times and quantities in orders / operations. In addition, it provides the ability to correct entered data later on.

Implementation considerations

You use the function package if you:

- would like to enter times and quantities for orders / operations so that these can be posted back to other systems, if required
- would like to modify data entered in the system at a later date.

Features

- Order administration
 - Configuration used to set customer-specific order numbers, order sequence numbers, operation numbers or split number lengths
 - Support for different types of orders like production orders, overhead cost orders, capacity orders, maintenance orders and project orders
 - Administrative function for locking or releasing operations
 - Administrative function for terminating operations that are not (no longer) logged on.
 - Administrative function for reactivating inadvertently logged off operations
- Order-related entry and posting functions:
 - Log-on /off, interrupt and partially confirm orders or operations
 - Entering yields, scrap quantities, rework quantities or open quantities
 - Posting of recorded production and downtimes to freely definable resource performance accounts
 - Comparing recorded workplace / machine times with the shift model stored at the workplace
 - Posting runtimes and numbers of units per operation
- Maintaining entered data using posting maintenance:
 - Displaying order-related log records (postings) generated based on the data entered
 - The ability to edit and correct order-related log records
 - Generating cancellation log records during processing after uploading to the ERP system
- Event maintenance
 - The ability to display and correct entered BDE events
 - The ability to edit and correct BDE events as an alternative to posting maintenance

- Recalculation function based on changed/corrected BDE events Generating updated log records (postings)
- Generating cancellation log records during processing after uploading to the ERP system

2 Fields of Application BDE

2.1 Overview

The goal of the following description is to present the many collection and processing options offered by HYDRA shop floor data collection (HYDRA-BDE) and machine data collection (HYDRA-MDE).

2.2 Basics of shop floor data collection

The postings of the HYDRA-side production data collection are based on the components machine, order and person. Depending on the approach and the main focus of the individual use of HYDRA, different components might play a prominent role or they might be unnecessary and therefore as far as possible disabled.

Machine/workplace – where does production happen?

Logging an order or employee on or off is basically machine or workplace related. Logging on to a machine answers the question of where a BDE posting takes place. BDE postings can basically be executed only on those machines or workplaces which have been configured accordingly in the HYDRA master data.

The names *machine* and *workplaces* are synonymous in HYDRA. In HYDRA-MDE, the expression "machine" is favored. In HYDRA-BDE, the name "workplaces" is favored.

Order/operation – what is produced?

All tasks performed by staff at machines are based on orders. Logging the order or its operation on answers the question what is done or which activity is executed.

HYDRA differentiates between different order types to classify orders according to their utilization. Different control information that is decisive for managing the orders is defined for each order type.

HYDRA provides the following order types by default. Further order types can be created according to the customer's requirements:

- Production order
A production order generally relates to the article/item and is characterized by a target quantity as well as a completion date.

- Overhead cost order
An overhead cost order (e.g. cleaning of the workplace) normally only refers to cost-accounting.
- Capacity order
A capacity order has only been designed for planning purposes (assignment of capacities) within HYDRA shop floor scheduling. You usually change capacity orders into production orders at a later stage.
- Project order
A project normally is unique. A project order combines the project steps that are carried out during a project.
- Maintenance order
A maintenance order has been designed for the planning or recording of maintenance measures.

In general, orders are created in a higher level system (e.g. ERP system) and transferred to HYDRA using an interface. Nevertheless, orders can also be created manually in HYDRA. Planning of orders (in the ERP system or HYDRA) depends on precise requirements.

Orders are mostly multi-level and divided into several operations to be processed on different workplaces and machines. The order and operation are generally logged on to the shop floor terminal at the individual workplace. Data can only be posted onto operations at shop floor terminals if a corresponding pool of orders exists in the HYDRA database.

The terms 'order' and 'operation' are mostly synonymous in the handbook descriptions. Usually, when posting functions on the terminal are being described, the operation is meant, and documentation on HYDRA shop floor data collection also emphasizes operation data. Only in certain HYDRA descriptions is the 'Order' used as an umbrella term for the whole multi-level production order.

Person - who is working?

The relation to the machine and operation is established in HYDRA by logging staff on and off at shop floor terminals. The work confirmations of employees form a basis for the calculation of personal expenses and for performance determination. With the logon or logoff of staff, you are informed who is working, where the person is working and what is done.

The relation of a posting to an employee is mostly optional. The logging on and off of production personnel is subject to fundamental decisions which are made before an MES system is introduced. HYDRA usually only allows those employees to log on, who are stored in the HR master data of the HYDRA database and authorized in the company.

2.3 Presentation of recording results

The components of data collection - machine, order, person, - are summarized to the greatest possible extent in logical dialogs. Consequently, an employee may, for example, log on and off to and from a terminal together with an order.

Decisions as to the design of individual dialogs on the shop floor terminals depend on the environment in which it is used, the industry sector, the organization, the range of items, the order structure and the machinery involved.

The data collection options in HYDRA represent a range of exemplary uses, which are supported by the modules of the standard system. A single installation requires only one selection of the possible configuration options. Individual HYDRA settings are finalized by the HYDRA user, usually together with an MPDV consultant, after successful installation and in the course of customizing.

2.3.1 Time recording

"Posting events" on shop floor terminals form the basis for the posting of particular machine times to an operation. A time posting is initiated by the posting event "logon" and completed logically by the posting event "logoff" or "interruption". This basic principle applies to all order and personnel posting events.

The time posting of shop floor data collection primarily occurs in two different time accounts, one for machine scheduling duration and the other for labor utilization.

The **machine time** is specified by the time interval between the logon and logoff of an operation. The machine scheduling time is harmonized with the shift calendar of the machine. Planned shift breaks are not included in the time interval calculation.

Personnel deployment represents the total of all labor times for each operation. This period is determined by the time interval between logon and logoff of the user or users. The basis for personnel postings is, once again, the shift model of the workplaces and the breaks it contains. If employees are processing more than one order simultaneously, then the HYDRA system carries out a proportional calculation of the operating time for each operation.

For further details on this please refer to the section entitled "posting of times".

The results are documented in "log records" that are generated automatically due to posting events.

2.3.2 Recording quantities

Quantities are separately maintained in HYDRA for each data collection level: machine, order, and person:

- **Machine related**
e.g. for display of the shift performance in the machine efficiency report
- **Operation related**
e.g. for display of the target-actual comparison in the order information
- **Personnel related**
e.g. as the basis for incentive pay based on quantities

Depending on the data collection type, quantities are either entered **automatically** (MDE/BDE terminal with counter connection) or must be **entered manually**.

Manual quantity postings are carried out in the entry dialog at the end of personnel or order processing or as partial confirmation/upload during order processing. Depending on the objective, the entry dialogs *Interrupt operation*, *Logoff operation*, *Logoff person* and *Partial confirmation/upload* are configured for the input of quantities. Quantities that are recorded manually are always posted onto the operation that is logged on. Posting onto the person realizing the posting depends on the posting type or the respective parameter settings configured in HYDRA.

Along with manual entry of quantities, HYDRA also supports **automatic entry of quantities** with active machine data collection. With this type of entry of quantities, a continuous quantity posting to the machine, and to the logged on operation and the logged on employees takes place at cyclical intervals.

All OPs and employees logged on to the machine are automatically assigned quantities. The quantities are computed according to the partitioning/cavity that is respectively defined for the operation and is posted onto the operations or persons.

HYDRA provides different accounts for quantity posting. The accounts that are predominantly used are yield and scrap. The two fields are provided appropriately during data collection. Scrap quantities can be classified by a scrap reason. The available scrap reasons are configured in HYDRA.

Other quantity accounts are *Rework* and *Open quantity*. These are usually only used specifically in form of customer-specific scenarios; the fields are provided as part of the customizing process.

2.3.3 Options of Machine Data Collection

If you connect a machine to a shop floor terminal, the automatic data collection is possible.

- Collection of production times and downtimes
- Direct assignment of malfunction reasons by collecting operating signals from machine control systems
- Piece number recording for yield, total quantity and scrap

Every workplaces and machine is assigned a shift model in the HYDRA configuration. The shift model determines the capacity of the machine and gives the shift calendar a reference time for the performance of a machine. All machines for which recording of standstills is carried out in HYDRA, have a definite machine status at all times during the shift. The total of all production and downtimes during a shift gives exactly the shift standard time.

Definition of downtime reasons

The determination of the relevant machine statuses forms the basis of machine monitoring by means of machine signals. Together with the "production" status all data collection relevant downtime reasons for the machine are created in HYDRA. The user may configure them individually.

Classification in HYDRA resource performance accounts (RPA)

HYDRA resource performance accounts are a system of time accounts, consisting of 12 accounts. They group together similar downtime reasons in a single account (e.g. all technical disturbances go in the "disturbance-caused interruptions" (DCI) resource performance account). During data collection, accumulating times are posted to the resource performance account to which the current machine status is assigned in the system configuration.

The standard definition of the HYDRA resource performance accounts is as follows:

No.	Acronyms	Description
1	SUT	Secondary utilization time
2	DCI	Disturbance-caused interruption (= technical interruption)
3	LCI	Logistics-caused interruption (= organizational interruption)
4	SCI	Staff caused (personnel related) <i>interruption</i>
5	IMN	Idle mode, not scheduled (e.g. repairs)
6	IMS	Idle mode, scheduled (e.g. maintenance)
7	SET	Setup
8	STA	Startup
9	U8	Free for user (e.g. trial production)
10	U9	Free for user
11	MUT	Main utilization time; "Production"
12	BKS	Neutral times, e.g. free, breaks etc.

If recording of downtimes is not required or if the recording of downtime reasons gives no clear result (e.g. for HYDRA group workplaces), then time recording generally takes place with *Production* status and is represented in the MUT main utilization time performance account.

The HYDRA resource performance accounts are also kept relating to operations and persons. Machine scheduling times (machine duration) for operations and personnel result from the following calculation:

Scheduling time = total (RPA 1 .. RPA 11).

Automatic recording of number of pieces using machine signal connection

Recording a cyclical signal from the machine enables the recognition of downtimes, and by counting the recorded cycles it is also possible to record the number of pieces produced.

To ensure an accurate determination of the number of pieces, HYDRA supports the multiplication of recorded cycles with multiple accesses per cycle. "Partitioning" (also called cavity) is a tool specific parameter, which is transferred to the terminal when an operation is logged on.

Automatically recorded data are posted as machine performance in HYDRA and simultaneously assigned to the operation and personnel currently logged on.

Collection	operation-related	Personal
actual quantities	X	X
machine times	X	X
Labor utilization	X	X

3 HYDRA-BDE Input Types (relating to operations)

3.1 Serial production

The serial production is the "classic" approach of manufacturing operations. This means that only one operation is produced at a machine at a time.

Serial production has the advantage that machine efficiency can be easily planned.

3.2 Parallel production

Parallel production means that more than one operation is processed simultaneously on the same machine.

At **individual workplaces** the system assumes that the two operations belong together (e.g. production of upper and lower parts with the same tool as two separated operations). Consequently, the system connects each person (who is logged on) with every operation. However, the labor utilization relating to operations is thereby halved.

At **group workplaces**, in contrast, parallel production is possible without the system linking the person to all operations that are currently being processed. This is mainly used at assembly workstations where several workers at the "Assembly" workstation process several operations in parallel. A disadvantage of this option is that the planning of such workstations is difficult.

3.3 Merged operations

This is a special type of serial production. At the planning stage (e.g. HYDRA shop floor scheduling) or at the shop floor terminal, different short running operations are grouped together in logical blocks with an easy to handle running time (i.e. merged operations). For these merged operations HYDRA creates a "representative" operation, which is logged on for all individual operations included. The entered data is divided according to different configurable perspectives.

3.4 Splits

If an operation should be processed on several machines in parallel, the HYDRA shop floor scheduling module allows for the operation to be split into several "splits". These splits are handled by HYDRA like separate operations and can separately be logged on and off independently of each other. The collection of all actual data then pertains to the particular splits and to the original operation (master split), which is known to the PPS system.

3.5 Multiple machine production

If no planning tool is used in HYDRA (e.g. HYDRA shop floor scheduling module), the system can be configured in a way that allows for an operation to be processed, i.e. logged on, on several machines at the same time, without having to be split. Data is collected separately for every machine. A disadvantage of this variant is that planning becomes very difficult.

3.6 "Mixed operation"

As regards the shop floor, HYDRA generally supports all forms of mixed operation. Consequently, splits, merged operations and "normal" operations can be processed serially, in parallel or as multi-machine production. Data collection refers in each case to the collection forms presented above.

In this way, it might be possible in HYDRA, for example, that while an upper and lower part are produced, the operation called "compression lower part" can be logged on to another workstation at the same time for reworking purposes (e.g. debur).

4 HYDRA-BDE Input Types (relating to personnel)

4.1 1.1 Summary

The sections that follow describe the different personnel-related views by way of examples. Postings based on people and on operations are displayed. To simplify matters, breaks are not included in the calculation of time intervals.

4.2 Single machine operation

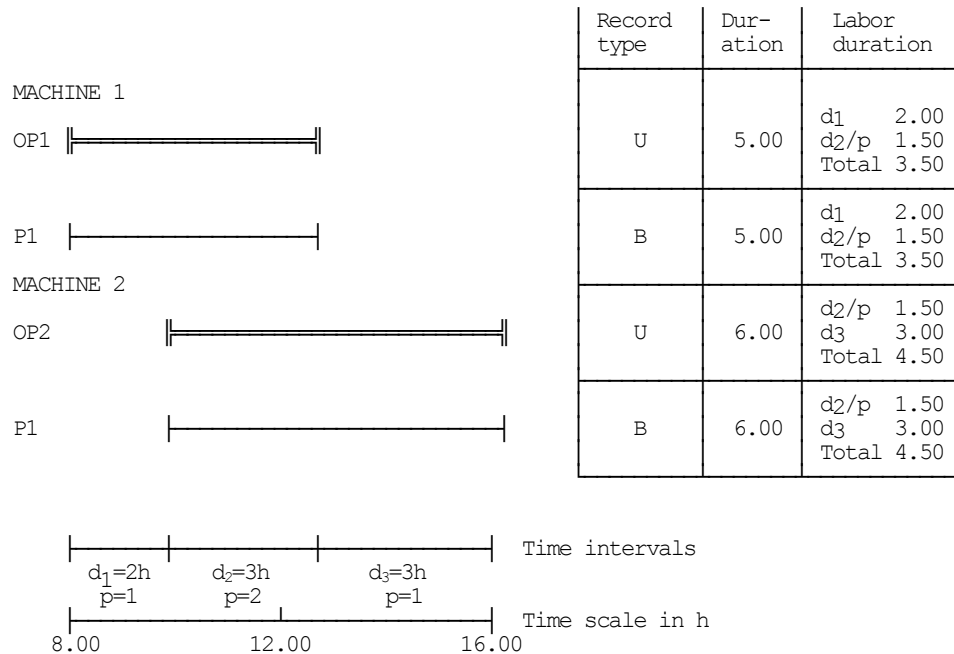
"Single machine operation" means that one person works alone on an operation at one workstation.

4.3 Multiple machine operation

Multi machine operation means that one person processes several operations at several workstations at the same time. This is normal, for example, on semi or fully automatic machines, where one person monitors several machines, feeding them with raw material and removing the finished parts. The labor utilization relating to operations is reduced according to the number of machines that are operated simultaneously.

Example

One person processes two operations at the same time on two machines.



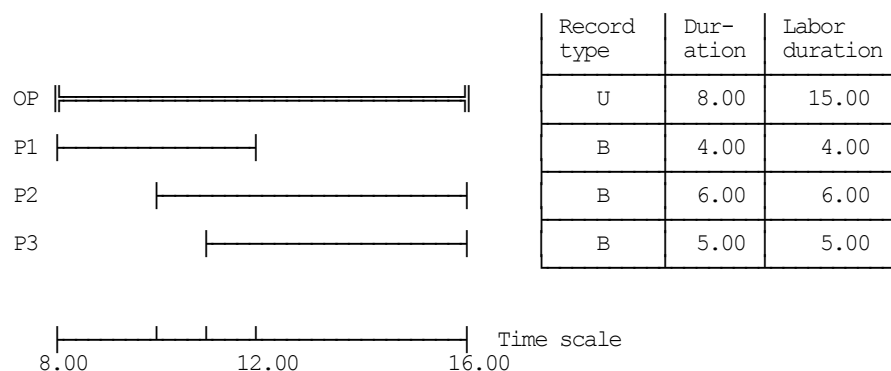
p : the number of times the same person has logged in
d_i: time interval i

4.4 Group work

Group work means that several people process a single operation at one workstation. With every additional person who is logged on, the labor utilization increases accordingly.

Example

Three people process an operation on one machine.



4.5 "Mixed operation"

HYDRA allows for all of the above-mentioned variants to be mixed, that is, HYDRA correctly handles the situation where a person with multi-machine operation is additionally logged on to a machine with group work.

Example

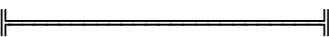
Several people process several operations on several machines (parallel production).

Machine 1 is an individual workplace

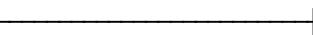
Machine 2 is a group workplace

Machine 1 Individual Workplace

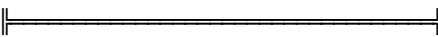
OP 1



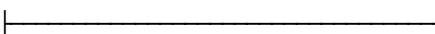
P1



OP2



P1



Record type	Dur- ation	Labor duration
U	d1 2	d1 2.0 P1
	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	Total 5	Total 3.2
B	d1 2	d1 2.0
	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	Total 5	Total 3.2
U	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	d4 1	d4/2 0.5 P1
	d5 2	d5/2 1.0 P1
	d6 1	d6 1.0 P1
	Total 7	Total 3.7
B	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	d4 1	d4/2 0.5
	d5 2	d5/2 1.0
	d6 1	d6 1.0
	Total 7	Total 3.7

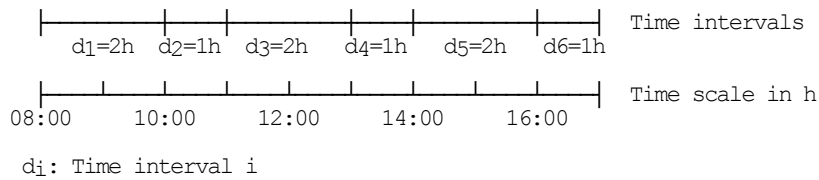
Machine 2 Group Workplace

OP 3

P2

OP 4

P1



U	d1	2	d1	2.0	P2
	d2	1	d2	1.0	P2
	d3	2	d3	2.0	P2
	d4	1	d4	1.0	P2
	Total	6	Total	6.0	
B	d1	2	d1	2.0	
	d2	1	d2	1.0	
	d3	2	d3	2.0	
	d4	1	d4	1.0	
	Total	6	Total	6.0	
U	d3	2	d3/3	0.7	P1
	d4	1	d4/2	0.5	P1
	d5	2	d5/2	1.0	P1
	Total	5	Total	2.2	
	Total	5	Total	2.2	

5 Data Collection in HYDRA-BDE

5.1 1.1 Summary

In HYDRA data is generally recorded by collecting and processing dialogs. Processing of a dialog results in events that are the basis for updating (calculation of statuses) and posting (generation of log records).

Core functions of data acquisition are the functions to resolve dialogs into events as well as checking and processing functions that lead to the generation of events within the system.

Dialogs

Dialogs represent the system's interface with the shop floor.

Events are generated by resolving the dialog. Resolving of the dialog into individual events is performed based on dialog data, the processing logic of the dialog (incl. configurations) and the context of the dialog (posting status).

Some of the HYDRA-BDE input dialogs are:

Dialog	Meaning
A_AN	Log operation on
A_P_AN	Log operation and person on (together)
A_TR	Confirm/upload OP partially
A_UN	Interrupt operation
A_AB	Log operation off
P_AN	Log person on
P_AB	Log person off
P_AAB	Log all persons off (1... n P_AB)
SA_AN	Log merged operation on
SA_TR	Partial upload of merged operation
SA_AB SA_ABME	Log merged operation off

Events

Events result from resolving dialogs and are thus the result of data input. Which events get to the system at what times, as a result, represents an important point of controlling the software.

Further, partly optional, data acquisition functions are:

- logging of the dialog and the resulting event,
- provision of the posting result,
- provision of current information in the form of lists (e.g. machine lists, order lists),
- escalations.

6 Validation Check

6.1 Overview

The validation check of a dialog verifies whether the system can completely process all resulting events (in the current status). If the dialog cannot be processed, i.e. not all events have been checked successfully, the dialog is rejected without making changes to the system.

The validation checks evaluate dialog data, the generated events and the HYDRA data set and interpret the posting status. Some (partly optional) configurations are also integrated.

Examples of **validation checks for dialog data**:

- if an OP is logged on, the logged on OP must exist in the HYDRA data set,
- if an OP is logged on, the machine where the OP is logged on must exist in the HYDRA data set,
- if staff is logged on, the logged on person must exist in the HYDRA data set,
- when postings are performed for group workplaces, the postings must include the persons.

Examples of **validation checks for the posting status**:

- if a part quantity is uploaded for an OP, the OP must be logged on to the machine where the posting is performed,
- if staff is logged off, the logged off person must have been logged on to the machine where the posting is performed.

Examples of **validation checks for configurations**:

- if an OP is logged on, the person who performs the logon must be authorized to log on the OP,
- if a part quantity is uploaded for an OP, the overdelivery quantity may not be exceeded.

Some validation checks are explained in more detail in the sections that follow:

6.2 Validation checks regarding overdelivery or underdelivery

6.2.1 Overview

If part quantities are uploaded for an operation or if an operation is interrupted or logged off, a validation check can be performed for the quantities posted. It is checked if there is an overdelivery. When an operation is logged off, you can also check if there is an underdelivery.

By default, quantity checks in HYDRA are designed for the manual input of quantities. The quantity check is performed if a posting dialog includes a quantity $\neq 0$. The quantity check is also always performed if a quantity $\neq 0$ is posted. In case of an overdelivery, the check is also performed if the quantity has already been confirmed and a deviation reason has been entered.

The overdelivery/underdelivery check, also referred to as target quantity check, can be activated for operations or for persons.

6.2.2 Operation-related activation of the overdelivery/underdelivery check

You can activate a validation check for overdelivery or underdelivery for the operation. The following values/settings are relevant:

Underdelivery

Value in percent. The posted quantity can deviate from the target quantity by the percentage specified.

Example:

Target quantity of the operation: 120 items

Underdelivery: 84%

The actual quantity must not fall below 101 items.

Overdelivery

Value in percent. The posted quantity can deviate from the target quantity by the percentage specified.

Example:

Target quantity of the operation: 120 items

Overdelivery: 168%

The actual quantity must not exceed 201 items.

Reaction

If the limits specified in the fields overdelivery or underdelivery are exceeded, a warning or an error message can be issued. Possible values are:

"blank" no reaction

W warning

X error

If a warning is activated ("W"), you can confirm a quantity deviation on the Windows terminal by entering a deviation reason. If an error is activated ("X"), a quantity deviation is rejected.

Notes

By default, the operation-related validation check is performed for the actual yield quantity recorded (primary quantity unit). You can also configure the processing code and activate a validation check for scrap, rework or problem quantities. A validation check for alternative quantity units (e.g. secondary quantity, etc.) is not provided in the standard.

In general: DOS terminals generally reject all quantity deviations with an error, even if the option is set to "W".

The functionality, which is available for Windows terminals, does not exist on the MOC.

If you use the SAP interface "PP-PDC", limit values for overdelivery/underdelivery are transferred as absolute values. HYDRA converts these values to percentage values in the interface.

6.2.3 Person-related activation of the overdelivery/underdelivery check

In addition or as an alternative, you can also activate a person-related overdelivery/underdelivery check. For this purpose, the HR master provides the "target quantity check" option in the "BDE" tab. Possible values:

- 1) No check.
- 2) Order logoff: When an operation is logged off, the system checks if the current yield is between the specified minimum and maximum target quantity. Both quantities are also specified in the HR master for a person.
- 3) Underdelivery/overdelivery: All quantity postings are checked for overdelivery, when orders are logged off/partially uploaded and interrupted. When an operation is logged off, it is also checked for underdelivery.

You can confirm the validation check on the Windows terminal, if you enter a deviation reason.

Notes

The person-related validation checks are only performed for yield (primary quantity unit).

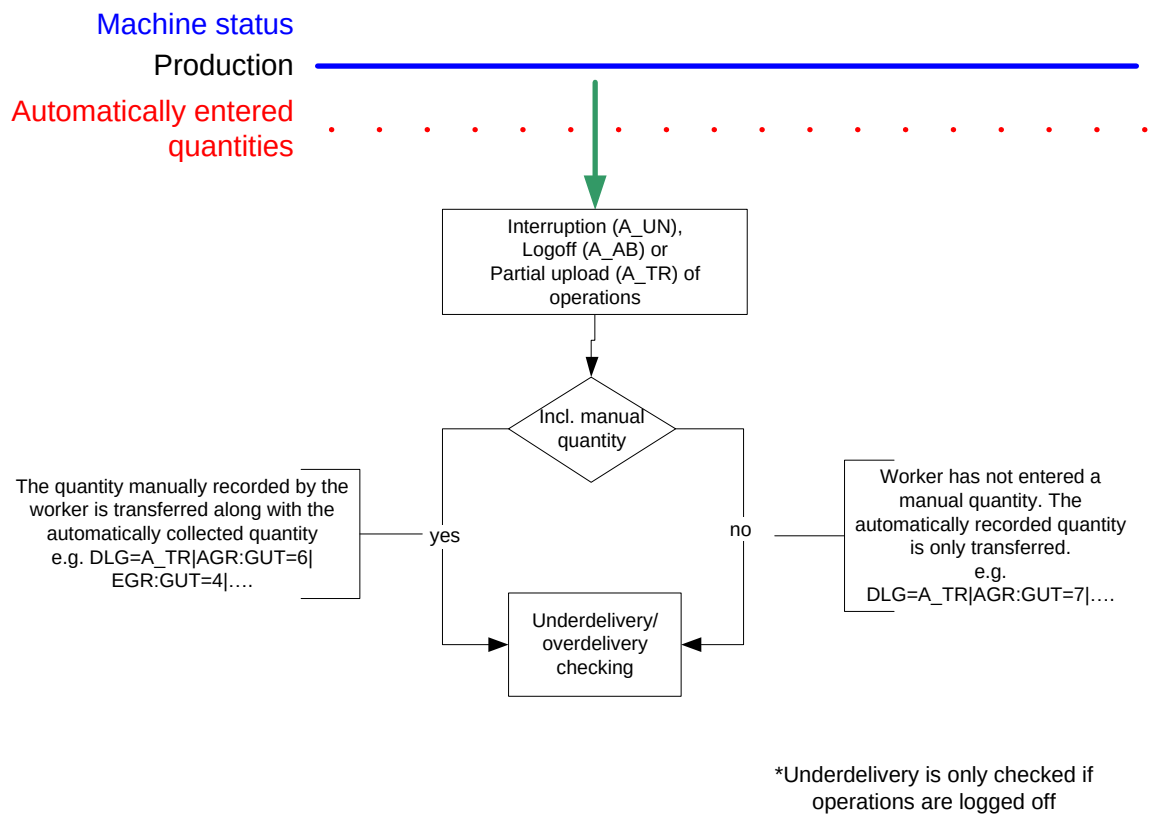
The limit values for overdelivery and underdelivery do not affect the person-related settings and are checked separately, if activated. If both validation checks are active, then first the person-related target quantity check is performed, then the operation-related check.

6.2.4 Overdelivery/underdelivery check with automatic recording of quantities

In HYDRA standard, the target quantity check is designed for the manual entry of quantities. It is only performed if a quantity $\neq 0$ is entered in the posting dialog. The paragraphs below describe different scenarios where the validation check is performed in connection with the automatic recording of quantities.

Scenario 1

If a quantity posting is performed for an operation in the active "production" status (interruption, logoff or partial confirmation/upload), the quantities that have been recorded automatically since the last posting (e.g. status change, operation posting, personnel posting) and the quantities that might have been recorded manually are sent to the server to be booked. A validation check is performed, because the command (DLG=...) sent to the server includes a quantity. It does not matter if the operator has manually entered a quantity $\neq 0$. The validation check is now performed for the quantity automatically recorded (AGR:GUT=...) and for the manually entered quantity (EGR:GUT=...=).



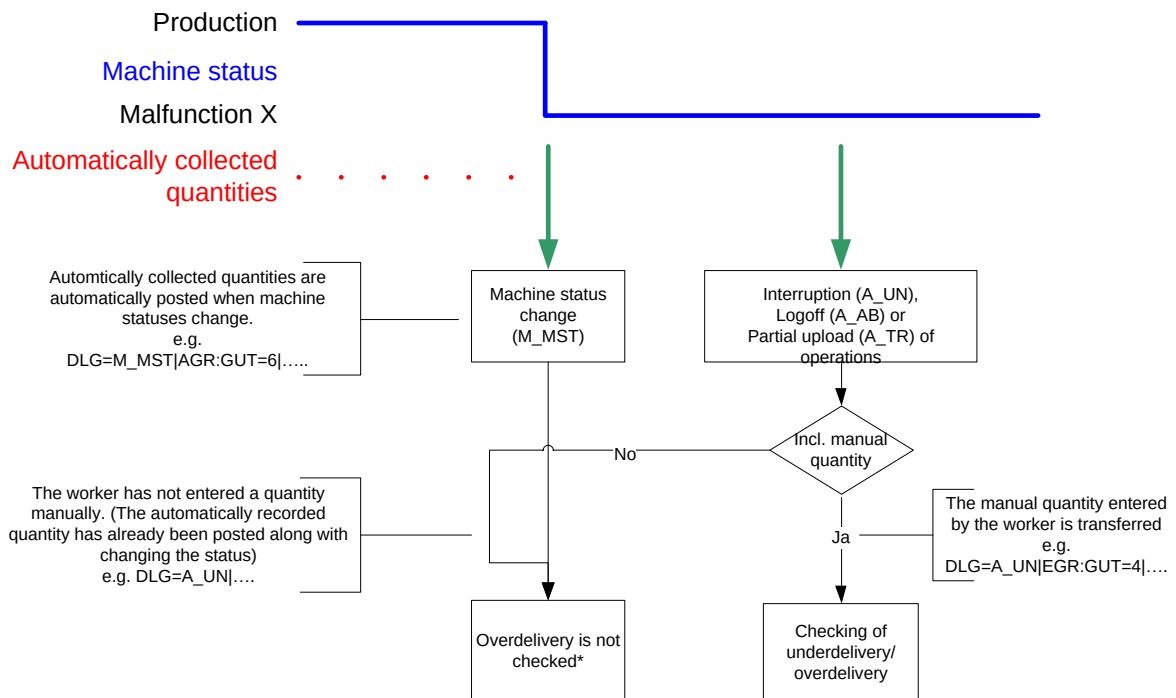
Scenario 2

If a machine switches from the production status to “malfunction”, the quantities recorded so far (AGR:GUT=...) are sent to the server with the status change (DLG=M_MST|...) to be booked. As this is an automatic posting (status change), a validation check is not performed.

If a posting is later on performed for the operation, a validation check is only performed if the operator manually enters a quantity $\neq 0$. This quantity is sent to the server with the posting (e.g. DLG=A_UN|EGR:GUT=...).

If the worker, for example, does not enter a quantity when the operation is interrupted (quantity = 0), no validation check is performed, because no quantity is sent to the server (DLG=A_UN|...).

As no automatic quantities have been recorded since the last status change, this posting does not include any automatic quantities, which could be checked.

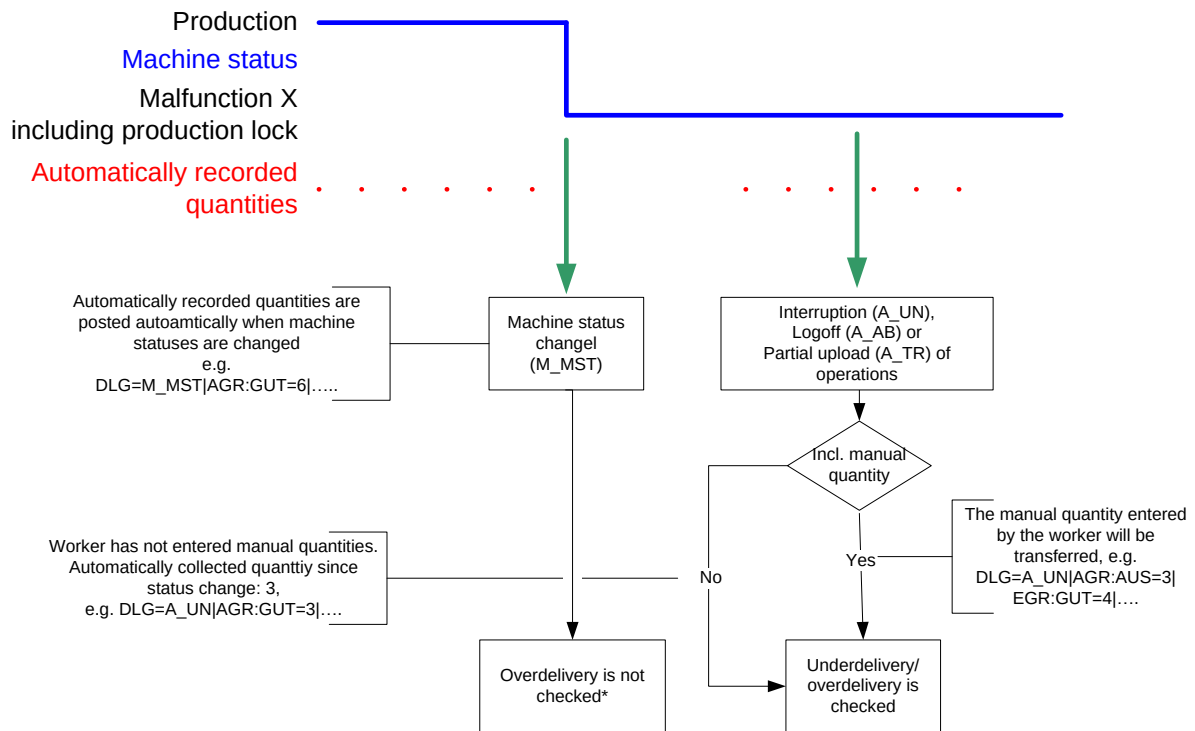


Scenario 3

Automatic quantities can still be recorded, even if the machine is in the malfunction status and the production lock is set.

In this case, this automatic quantity is sent to the server along with the manual quantity entry (e.g. DLG=A_UN|AGR:GUT=...) and is checked for validity.

The validation check is performed as described in scenario 1. It does not matter if the operator has made manual quantity entries or not.



7 BDE Log Records

7.1 General

The "posting" function generates log records from recorded events. Log records describe a period of time and include the evaluation of different values, such as quantities, durations or performances.

A BDE log record is an evaluated data record that is generated due to posting events. "Evaluated" means that this data record

- generally refers to a period of time (in an extreme case, this period of time is reduced to a point in time),

and

- includes durations (machine times, labor utilization, resource performance accounts) or
- evaluated quantities (e.g. yield, scrap)

HYDRA-BDE log records describe postings based on operations, staff and batches. Which log record is generated at which point in time depends on different factors, such as triggering posting events, configured posting rules or the HYDRA modules in use. The different types of log records are differentiated by their record type. The following record types based on orders are distinguished:

7.2 Record type A

General

A log record of the record type A is generated when an operation is logged on. It has been designed to be able to upload operation logons to the higher-level ERP/PPS system.

Triggering events and dialogs

Events: Logon of operation (A_AN)

Dialogs: Log operation on (A_AN),

Log operation and person on (together) (A_P_AN),

Log operation on with output batch (A_AN_MPL),

Beginning of shift (A_AAN)

etc.

Special remarks

Using optional configurations, different other dialogs can also cause an operation to be logged on (e.g. change order if the machine status changes).

The log record is not displayed in the maintenance of postings function. Consequently, it cannot be edited (changed, deleted).

7.3 Record type T

General

A log record of the record type T is generated when "partial uploads/confirmations" are recorded. In contrast to all other record types that refer to a period of time, this log record always refers to exactly one point in time (beginning of posting and end of posting are identical in the log record). Thus, this log record does not include durations, i.e. fields relating to durations do not include a value (0). The log record contains the quantities recorded with the upload.

Triggering events and dialogs

Events: all events that include a manual quantity (A_TR, A_UN, A_AB, etc.)

Dialogs: all dialogs that include a manual quantity .

Special remarks

In general, a partial upload may be performed for an operation that is logged on. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded by way of a "quantity upload".

All quantities posted onto an operation are included in the single partial uploads (record type T) and as total in the log records "interruption of operation" (record type U) and "logoff of operation" (record type E). Thus, a T record is generated for each quantity that is collected and posted on an operation.

Manual quantities may be recorded in any dialogs, e.g. in a partial upload, when personnel is logged off, when an operation is interrupted or logged off, when batches are posted, etc. The log record of the record type T with manual quantities is directly generated, when the dialog is recorded/posted (not only when the operation is logged off/interrupted, for example).

In addition, the automatic collection of counters allows for quantities to be posted onto active operations. T records are generated for these quantities as well. HYDRA accumulates the quantities from the automatic collection over a period of time and generates the corresponding T record(s), **but only if the scenario changes**. This may be, for example, an order or person that is logged on manually, a manual, partial upload/confirmation or a manual or automatic machine status change.

The single partial uploads/confirmations also represent the collection of quantities with different reasons (e.g. scrap reasons).

Example:

Scrap (quantity 1) with the reasons 1, 2 and 3 each was recorded for an operation. The U or E record shows a total scrap quantity of 3. In addition, HYDRA records three T records each with a scrap quantity of 1 and the corresponding reason.

7.4 Record type U

General

A log record of the record type U is generated, when an operation is interrupted. Either the user can interrupt the OP manually or it can also be made automatically, for example, by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record of an operation includes the period of time between logon and interruption of the operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the operation was logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities and times posted in the log record are also posted onto the operation status.

Triggering events and dialogs

Events: Interruption of operation (A_UN),

Quantity upload to an inactive operation (A_MR)

Dialogs: Interruption of operation (A_UN),

Quantity upload to an inactive operation (A_MR)

End of shift (A_AUN)

etc.

Special remarks

In general, an operation may only be interrupted if it was logged on before. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded using a "quantity upload". In this case, the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be interrupted even by different other dialogs through optional configurations (e.g. order change when the machine status is changed or automatic interruption of the operation when the target quantity is reached).

7.5 Record type E

General

A log record of the record type E is generated, when an operation is logged off. As regards content, this data record corresponds to a log record of the record type U.

Triggering events and dialogs

Events: Operation is logged off (A_AB),

An inactive operation is terminated (A_BE)

Dialogs: Operation is logged off (A_AB),

An inactive operation is terminated (A_BE)

etc.

Special remarks

If an inactive operation is finished the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. the operation is logged off automatically, when the target quantity is reached or predecessor operations are logged off automatically).

7.6 Record type H

General

A log record of the record type H is generated, when an (output) batch is changed/logged off (ADE-CHV, MPL). Either the user can change the batch manually or it can also be made automatically, for example, by a machine signal or by interrupting the operation or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between logon and logoff of the batch to/from the machine and operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the batch and operation were logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities posted in the log record are also posted onto the batch.

Triggering events and dialogs

Events: Logoff of (output) batch (CA AB)

Dialogs: (Output) batch change (CA_WL)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN) etc.

Special remarks

A batch posting does neither include labor utilization nor personal resource performance accounts.

A batch may also be triggered to be posted even by different other dialogs through optional configurations.

7.7 Record type B

General

A log record of the record type B is generated if a person is logged off from a workplace and an operation that is active at this workstation. Either the user can log the person off manually or it can also be made automatically, for example, by interrupting the operation or by the person clocking out (HYDRA-PZE: out) or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between the person being logged on and logged off to and from the workplace and operation. The durations included in the log record refer to exactly this period of time and, subject to the respective configuration, they are compared with the shift model of the workplace to which the person and operation were logged on or with the person's BDE shift model: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

Triggering events and dialogs

Events: Log person off (P_AB)

Dialogs: Person is logged off (P_AB)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN)
 HYDRA-PZE out (P_GEH)
 etc.

Special remarks

A person may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. staff is logged off when the machine status is changed).

7.8 Special features

Evaluation date

The field "evaluation date" is defined for the work day evaluation of PZW (Personnel Time Management) and the wage calculation of the LLE module. It has been designed to assign postings to a settlement day for the LLE module and for the BDE/PZW comparison. Due to flexible working times, this settlement day may differ from the ADE shift date, in particular for night shifts.

Further information can be found in the document entitled [GLOSSARY_EvaluationDate.pdf](#).

Corresponding shift

The time stamp for logging off pertaining to a BDE log record determines the shift (shift date, shift number) to which the log record belongs.

Operation splits

When it comes to split operations, the log record is ONLY generated for the split operation. A log record is not generated for the split master itself. The status of the split master is only updated.

Merged operations

Log records are generated for merged operations as well as for individual operations. The statuses of individual operations are also updated accordingly. While posting, the recorded quantities and times of a merged operation are distributed to the corresponding individual operations according to different configurations. The log records of individual operations only include proportionate quantities and times.

8 Updating of the Posting Status

8.1 1.1 Summary

The functional area "updating of the posting status" has been designed to calculate statuses on the basis of recorded events. A status describes the condition of an object regarding the values that vary over time (e.g. OP status, posting status, etc.) and the relations between objects (e.g. OP is logged on to machine, person is logged on to OP and machine).

Usually, posting relations are generated for logons (operation to machine, person to machine and operation) and canceled when they are interrupted/logged off.

An object's posting status is updated every time when the event to be posted directly or indirectly affects the object.

Examples:

- By logging an operation on, this registered operation is set to the "running" status. The "operation logon" event directly affects the operation.
- By way of automatically recorded machine counters, resulting quantities are posted onto the operations that are currently active at the machine. The "automatic counter posting" event indirectly affects the operation(s).

The posting status also has an essential influence on which events are triggered by a dialog. Thus, the "log all persons off" dialog (P_AAB), for example, determines all persons who are active at the machine from the posting status and generates or posts a "personnel logoff" event for each person.

9 Posting of Times

9.1 General

The machine duration (or just duration) is determined by the time interval between logon and logoff of an operation. The machine scheduling time is harmonized with the shift calendar of the machine, whereas planned shift breaks are removed from the time interval calculation.

The labor utilization (or labor duration) is the sum of all labor times for each operation. This period is determined by the time interval between logging the user(s) on and off. The shift model of the workplace and the breaks included in this model are the basis even for personal postings (record type B). If several persons process several orders at the same time, the working time will be allocated proportionately for the corresponding operation in HYDRA according to the number of operations to which the persons are logged on.

The determined times (duration, labor utilization) are updated in the operation status (actual duration, actual labor utilization) and saved within the posting results - the log records.

Within the log record, assignment and meaning of time-related fields depend on the record type of the log record:

Field	Record type U/E	Record type H *)	Record type B
Duration	The operation's production time on the workplace	The output batch's production time on the workplace	The person's sheer registration time on the operation.
Labor utilization	Total of labor utilization of all persons logged on to the operation.	No labor utilization is defined in the log record for batch postings.	The person's labor utilization at the operation = evaluated registration time depending on the person being logged on (number of operations and workplaces).
RPA based on orders	The operation's production time on the workplace	The output batch's production time on the workplace	The person's sheer registration time on the

Field	Record type U/E	Record type H *)	Record type B
	distributed onto resource performance accounts. Please note: Subject to the current machine status, the resource performance account, which has been assigned to this status during the configuration, is "charged".	distributed onto resource performance accounts.	operation distributed onto resource performance accounts.
Personal RPA	Total of the personal resource performance accounts (see record type B) of all employees logged on to the operation.	Personal resource performance accounts are not defined in the log record for batch postings.	By way of the "proportionate RPA posting in personnel postings" option, the person's resource performance accounts (just as it is the case for the labor utilization) can be calculated proportionately - depending on the person's registration (number of operations and workplaces). If this option is not active personal RPAs correspond to RPAs based on orders.

*) Batch posting (only with operations that are subject to management in batches).

Time-related fields do not play a role with log records of the record type A and T (always 0) and therefore they are not included in this overview.

The document entitled "data collection in HYDRA-BDE (personal)" explains the process of determining times within personnel postings in different scenarios using examples.

Please note:

The determination of durations when status postings are recorded online at the terminal (or via PDM) only takes into account a limited number of shifts relating to the point in time of the last posting made for this machine.

However, durations might be missing or faulty in BDE log records if no postings (order postings, personal postings, status changes, shift changes, or similar) are entered for a machine over a longer period of time (e.g. the terminal is shut down).

9.2 Special features

Logging staff off during status 30000 "not assigned"

If a person logs off in status 30000 "not assigned", the duration of the labor utilization will be posted onto the resource performance account that is assigned to the "general disturbance" status within the personal BDE log record (record type "B"). If then the machine status is changed, the machine is assigned this new status over the complete period of time. Thus, resource performance accounts might vary.

Specific configurations

The following configuration options may influence the above-mentioned posting of times/durations:

Proportionate posting for parallel OPs

Configuration based on workplaces (tab configuration > general).

Project time recording

When it comes to project time recording, the times to be posted are recorded manually instead of being calculated by the system.

Post production time to main utilization time (MUT) during break

Cross-system configuration in the basic parameter settings of HYDRA.

RPA to calculate labor times

Cross-system configuration in the basic parameter settings of HYDRA.

Proportionate RPA posting in personnel postings

Cross-system configuration in the basic parameter settings of HYDRA.

Synchronize labor utilization with the person's HYDRA BDE shift model

Cross-system configuration in the basic parameter settings of HYDRA.

HYDRA-ADE waiting period processing

Cross-system configuration in the basic parameter settings of HYDRA.

Merged operations

If several operations are grouped into a merged operation, the times are distributed onto the individual operations according to different configurations.

Split operations

Further information on how to post split operations or their split master can be found [here](#).

10 Posting of Quantities

10.1 General

Manually collected quantities

Manual quantities are always posted onto the registered operation and, subject to the respective configurations in HYDRA, to the person carrying out the posting as well. The below table shows whether or when manually collected quantities are posted onto the operation and/or the reporting person.

Posting	Posting to operation	Posting to reporting person
Interrupt OP	Yes	No *) **)
Log OP off	Yes	No *) **)
Partial confirmation/upload	Yes	Yes **)
Log person off	Yes	Yes

*) compare the "Notes on personal posting of quantities"

**) The person who performs the posting needs to enter his/her personal badge number in the posting dialog.

Please note for entering the quantity when personnel is logged off:

If several operations are logged on simultaneously, the quantities will be posted onto all operations.

Recommendation:

The relevant quantities based on operations should be recorded as partial upload for each operation, before the person logs off (without entering a quantity).

Automatically collected quantities

All OPs and employees logged on to the machine are assigned automatic quantities. Automatic quantities result, for example from counter collection.

The quantities to be posted are computed according to the partitioning/cavity that is respectively defined for the operation and operations or staff are posted.

Notes on the personal posting of quantities

The personal posting of quantities depends on the workplace on which an operation was produced. In general: automatically collected quantities are always posted onto all persons who are logged on.

Individual workplace (EAP)

If quantities are neither recorded automatically nor with respect to staff (i.e. no quantity is entered when a person logs off), no quantities will be taken over to the person. Consequently, the quantity fields within the personal posting dialog (posting of record type "B") are empty (0).

Exceptions:

- a) The *"quantity posting to staff"* configuration option within the machine/workplace configuration allows for the manually recorded quantity to be posted onto the person who has so far been logged on for the longest time, when an OP is interrupted or logged off.
- b) Using the option of the HYDRA basic parameter settings "post manual quantities as automatic quantities" allows for manually recorded quantities to be posted like automatically collected quantities, i.e. they are posted onto all OPs and persons logged on.

Group workplace (GWP/GAP)

At group workplaces, the quantity is always taken over to the person when the OP is interrupted or finished, as a unique assignment between person and operation has been established.

10.2 Special features

The following configuration options may influence the above-mentioned quantity posting:

Post manual quantities as automatic quantities

Cross-system configuration in the basic parameter settings of HYDRA.

Quantity posting to staff

Configuration based on workplaces.

Merged operations

If several operations are grouped into a merged operation, quantities are distributed onto the individual operations according to different configurations.

Split operations

Further information on how to post split operations or their split master can be found [here](#).

11 Controlling the Sequencing List

General

The sequencing list of the shop floor terminal is flexible and can be adjusted to meet specific requirements. The settings described below control how data is provided in the sequencing list. Please keep in mind that certain settings can only be made during the customizing process.

Workplace/ machine configuration: "Sequencing list" setting

Description

Which order pool is used to generate the sequencing list can be defined in the *sequencing list* configuration of the [workplace configuration](#) dialog.

The pool of orders is determined by planning in HYDRA shop floor scheduling or by the specifications from the PPS system and is a result of whether the OP was planned directly for a machine or a machine group.

Possible settings

S - Basic setting

The value is transferred from the option with the same name in the [HYDRA basic parameter settings](#).

M - Pool of workplaces

Only the operations planned for the workplace are displayed in the sequencing list at the terminal.

G - Pool of workplaces and groups

The sequencing list at the terminal displays those OPs that are either planned for the current workplace or for a different workplace in the group or that are still in the pool of groups.

K - Pool of workplaces and categories

The sequencing list at the terminal displays only the operations that are planned for workplaces in the same machine category.

H - Group control

The sequencing list at the terminal displays those OPs that are either planned for the current workplace or for a different workplace in the group.

Workplace/ machine configuration: "Number of OPs" setting

Description

The maximum number of OPs for the sequencing list can be configured by the option "number of OPs" within the [workplace configuration](#). The OPs are selected in ascending order based on how the sequencing list is sorted.

Possible settings

0 = no restriction (display all existing OPs)

1-999 = maximum number of OPs

Order types: "Sequencing list" setting

Description

The configuration is made at order level in the *sequencing list* option under [order types](#).

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- F The OP should only be displayed in the sequencing list if it is fixed.
- N The OP should not be displayed in the sequencing list.

Processing code: "Sequencing list" setting

Description

Whether or not an OP with this processing code should be displayed in the sequencing list can also be defined in the [processing codes](#) `..\..\functions\moc\MOC_ProcessingCodes.pdf` configuration.

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- N The OP should not be displayed in the sequencing list.

Order status assignment: "Sequencing list" setting

Description

Whether an OP should appear in the sequencing list based on its status can be defined using this option under [order status assignment](#). If an "N" is defined here, an OP in this status is not displayed in the sequencing list.

This is used as a standard feature to ensure that only prepared or interrupted OPs appear in the sequencing list.

Furthermore, it can be configured that running OPs also appear in the sequencing list.

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- N The OP should not be displayed in the sequencing list.

Miscellaneous notes

Generally, the following operations never appear in the sequencing list:

- Locked operations
- Individual operations from merged operations that were generated at MOC.
- The original operation for split operations
- Operations of inactive alternative sequences

Order of operations in the sequencing list

Generally, the data listed below defines how the operations are ordered in the sequencing list at the terminal:

1. Internal field for the planned start date (auftrags_bestand.sort_dat)
2. Internal field for the planned start time (auftrags_bestand.sort_dat)
3. HYDRA combined order/ operation number (auftrags_bestand.auftrag_nr)

The system fills in the internal fields for the planned start date and planned start time. This depends on how the operation was created in the system (transferred via the interface or manually created in the system) and on the planning functions used subsequently. How the system does this is described in each applicable documentation.

- [Manual editing via the user interface or via the system interface – initial data creation](#)
- [Manual editing via the user interface or via the system interface - modification](#)
- [Using HYDRA Shop Floor Scheduling](#)
- [Using HYDRA Order Sequencing](#)



When using order sequencing, we recommend to only use the "M" option (pool of workplaces) to configure the sequencing list; the order within the sequencing list at the terminal is not defined when the other options are used.

12 Order and Operation related Functions

Purpose

You can use the functions described below to perform activities for orders or operations.

Integration

You can call the functions from several applications. The relevant applications are listed below in the descriptions of the different functions.

Requirements

The orders/operations must be created in the system. The function descriptions include the requirements to use the separate functions.

Change order status

Function authorization	or.statchg	Change order status
------------------------	------------	---------------------

The status of a complete order can be changed using this function. The function is integrated in the following application:

- Order overview → Tab *Order* → Category *Change order status*

Processing using BDE 8.1 and BDE 8.2 including Service Pack 14 (only applies when using HYDRA):

When you call this function, the dialog described below is called for each order selected.

Processing using BDE 8.1 and BDE 8.2 including Service Pack 15 (only applies when using HYDRA):

If exactly one order is selected, the dialog *Change order status* described below is opened when you call this function. The dialog provides the following fields

Order

Order number

Current status

Current status of the order

Action

The current status controls the available actions:

Current status	Possible new statuses
Not free	- Release order (default)

Current status	Possible new statuses
(specific status that is only available as part of a customization)	- Set order to "in process" - Terminate order
Prepared	- Set order to "in process" (default) - Terminate order
Started	- Terminate order

With the other statuses, you cannot select any action.

If several orders are selected, the processing is as follows:

- If all orders selected have the same order type and the control indicator "S" (other status), then the orders are set to the status with control indicator "V" (prepared).
- If all orders selected have the same order type and the control indicator "V" (prepared), then the orders are set to the status with control indicator "L" (in process).
- If all orders selected have the same order type and the control indicator "L" (in process), then the orders are set to the status with control indicator "E" (finished).
- If the orders selected have a status with a control indicator that is not "S", "V" or "L", then the button is disabled.
- If the orders selected have different statuses, then the button is disabled.

You can only set the order status to finished if no operation is currently logged on. If the order status is set to "finished", the status is also set to "finished" for all operations of the order.



If the order is finished using this function, the system only generates a log record (record type E) for the operations of this order if the option *Generation of log record terminate OP (client)* is set in the basic settings. To generate the log record, the workplace is used that is stored for the operation (it does not matter if the operation is planned for the workplace or not). If no workplace is stored, the workplace with the lowest number of the group is identified.

Update order

Function authorization	or.actualize	Update order
------------------------	--------------	--------------

Use the function *Update order* to update one or several selected orders. The action "update order" is directly performed. No window opens to confirm the action. Via the update function, not only data is requested, but the recalculation of values (formulas) is also started.

After the update, the order data is refreshed, so that also the data display is updated.

The following applications provide this function:

- Order overview → Tab *Order* → Category *Execute*
- Order information → Tab *Main Page* → Category *Execute*

- Edit operations → Tab *Other functions* → Category *Change status*

Schedule order

Function authorization	or.terminate	Schedule order
------------------------	--------------	----------------

Use the function *Schedule order* to schedule one or several selected orders. The action "schedule order" is directly performed. No window opens to confirm the action.

You can also call the scheduling for one or several selected operations. In this case, the orders of these operations are scheduled.

After the update, the order data is refreshed, so that also the data display is updated.

The following applications provide this function:

- Order overview → Tab *Order* → Category *Execute*
- Order information → Tab *Main Page* → Category *Execute*

Change operation status

Function authorization	op.statchg	Change operation status
------------------------	------------	-------------------------

Use this function to change the status of an operation. Whether you can/may change the status of an operation depends on its current (previous) status.

You cannot change operations with status *Running* using the function *Change status*. You can only change these OPs using *Interrupt OP* or *Log OP off*. You can only set an interrupted operation to *Running* using the function *Log OP on*.

The following only applies when using HYDRA and HYDRA BDE:

Processing with BDE 8.1 or BDE 8.2 including Service Pack 14

When you call this function, the dialog *Change operation status* described below is called for each operation selected.

Processing with BDE 8.2 as of Service Pack 15:

If exactly one operation is selected, the dialog *Change operation status* is opened when you call this function. The dialog provides the following fields

Order / Sequence / OP / Split

Operation number

Current status

Current status of operation

Action

The current status controls the available actions:

Current status	Possible new statuses
Not free (specific status that is only available as part of a customization)	- Release (default) - Terminate
Prepared	- Terminate
Interrupted	- Terminate

With the other statuses, you cannot select any action.

If several operations are selected, the processing is as follows:

- If all operations selected have the same order type and the control indicator "S" (other status), then the operations are set to the status with control indicator "V" (prepared).
- If all operations selected have the same order type and the control indicator "V" (prepared) or "U" (interrupted), then the operations are set to the status with control indicator "E" (finished).
- If the operations selected have a status with a control indicator that is not "S", "V" or "U", then the button is disabled.
- If the operations selected have different statuses, then the button is disabled.

Depending on the option *Generation of log record terminate OP* in the basic system settings, the system not only sets the operation to status "E", but also generates an end posting (log record of record type "E") that can be uploaded to the higher-level ERP system when you finish an operation. When an order is finished, all operations are finished.

To generate the log record, the workplace is used that is stored for the operation (it does not matter if the operation is planned for the workplace or not). If no workplace is stored, the workplace with the lowest number of the group is identified.

You can also use this function to finish overhead cost operations that cannot be finished using the "normal" posting dialogs A_AB on the terminal and the console.

You can change the status of a waiting period operation via *Change status*. The status change is not important because during a waiting period operation no activity is performed. The status change is used to inform the user. The posting functions in the dialog boxes are not relevant for waiting period operations. Also error messages are not relevant that you might receive with logon or logoff.

Change secondary status

Function authorization	op.secstatchg	Change secondary status
------------------------	---------------	-------------------------

Use the button *Change secondary status* to change the secondary status of an operation. You can only call the function if the current status of the operation is configured with the option "Editable secondary status" (customization).

The function is integrated in the following application:

- Order overview → Tab *Operation* → Category *Change status*

The following only applies when using HYDRA and HYDRA BDE:

Processing with BDE 8.1 or BDE 8.2 including Service Pack 14

When you call this function, the dialog *Secondary status* is called for each operation selected where the new secondary status can be set.

Processing with BDE 8.2 as of Service Pack 15:

If exactly one operation is selected, the processing is as follows:

- The dialog *Secondary status* is called. The dialog displays the order, operation and split number for the selected operation and the text of the previous secondary status.
- Use the combo box to select a new secondary status.
- Confirm ✓ to set the new secondary status for the selected operation.

If several operations of the same order type are selected, the processing is as follows:

- The dialog *Secondary status* is called. The dialog only provides the combo box to set a new secondary status.
- Use the combo box to select a new secondary status.
- Confirm ✓ to set the new secondary status for all selected operations.

You can manually enter the new secondary status or select the status from a selection list of secondary statuses. Note: The field is a mandatory field. You cannot delete the current status. If you want to "reset" the set secondary status, a respective secondary status must be provided.

Change resource status

Function authorization	op.resstatchg	Change resource status
------------------------	---------------	------------------------

Use the function *Change resource status* to change the resource status of an operation. The following applications provide this function:

- Order overview → Tab *Operation* → Category *Change status*
- List of Material Requirements

The following only applies when using HYDRA and HYDRA BDE:

Processing with BDE 8.1 or BDE 8.2 including Service Pack 14

When you call this function, the dialog *Change resource status* is called for each operation selected where the new resource statuses can be set.



All three resource statuses are always set at the same time.

Processing with BDE 8.2 as of Service Pack 15:

If exactly one operation is selected, the processing is as follows:

- The dialog *Change resource status* is called. The dialog displays the order, operation and split number of the selected operation and the preceding resource statuses.
- Enter new values in the input fields *Person OK*, *Tool OK* or *Material OK*.
- Confirm ✓ to set the new resource status for the selected operation.



All three resource statuses are always set at the same time.

If several operations of the same order type are selected, the processing is as follows:

- The dialog *Change resource status* is called. The dialog only provides the fields *Person OK*, *Tool OK* and *Material OK*.
- Enter new values in the input fields *Person OK*, *Tool OK* or *Material OK*.
- Confirm ✓ to set the new resource status for all selected operations.



All three resource statuses are always set at the same time.

Lock operation

Function authorization	op.lock	Block operation
------------------------	---------	-----------------

Use the function *Block* to lock one or several selected operations. You cannot make BDE postings for a locked operation and the locked operation is not shown in the sequencing list on the terminal (applies only when using HYDRA).



An operation lock does not affect the planning in the shop floor scheduling.

The following applications provide this function:

- Order overview → Tab *Operation* → Category *Lock*
- Edit operations → Tab *Other functions* → Category *Lock*

The action is directly performed. No window opens to confirm the action. If at least one operation of the selected operations is already locked, the lock is also performed. No error message is shown for the operation already locked. An error message is only shown if an operation cannot be locked that has not been locked before.

Unlock operation

Function authorization	op.unlock	Unlock operation
------------------------	-----------	------------------

Use the function *Unlock* to unlock one or several selected operations. The operation can again be logged on and is again displayed in the sequencing list.

The following applications provide this function:

- Order overview → Tab *Operation* → Category *Lock*
- Edit operations → Tab *Other functions* → Category *Lock*

The action is directly performed. No window opens to confirm the action. If at least one operation of the selected operations is not locked, the unlock is also performed. No error message is shown for the operation not locked.

An error message is only shown if an operation cannot be unlocked that has been locked.

Note: Operations that have been split (the so-called split master) are also locked. You cannot use this function to unlock these operations!

Reactivate operation

Function authorization	op.reactivate	Reactivate operation
------------------------	---------------	----------------------

Use the function *Reactivate* to reactivate finished operations in the system. The function is integrated in the following application:

- Order overview

The following only applies when using HYDRA and HYDRA BDE:

Processing with BDE 8.1 or BDE 8.2 including Service Pack 14:

You can only call the function for exactly one operation. If several operations are selected, the function is not active. You can only call this function for finished operations.

When the user calls the function, the user must confirm the reactivation in a dialog. Confirm to reactivate the finished operation: the operation changes to the status with control indicator *interrupted*. You can now log on the operation again.

Processing with BDE 8.2 as of Service Pack 15:

If exactly one finished operation is selected, the processing is as described above.

If several finished operations are selected, the selected operations are reactivated. No dialog is displayed.

13 Day Types

Overview

HYDRA menu	Master data → Production control → Day types
FEDRA menu	Detailed scheduling → Master data → Day types
Transaction code	dtmf
Function authorization	mddtmf

Day types are defined to specify the shift schedule for every workday. In a day type, you specify the planned working times and break times for up to four shifts a day.

Purpose

You should consider the following issues prior to the definition and configuration:



You only need one day model for days with identical shifts.



Several (different) day models are required, if the hours worked vary (e.g. "early Friday").



Machines with the same shifts but different breaks require different day models.



Machines having the same shifts, but a different number of shifts (2-shift operation, 3-shift operation) require different day models.

Only the last shift of a day model may include a day changeover/midnight (e.g. 10:00 p.m. – 6:00 a.m.)!



If a day type is changed, this change *directly* affects all machines where this day model is assigned in the year model. Therefore, we advise against changing day models that are currently being used (i.e. changes affecting that specific day).



If, contrary to expectations, modifications are required these modifications may exclusively refer to subsequent shifts (not to the current shift). Furthermore, these modifications must be carried out at least 2 hours before the change of shifts. Reason: Different system components read the

shift models in cyclic intervals and compare them with the locally saved shifts.

Requirements

The days off must be specified beforehand.

Selection criteria

Day type ... until ...

Restricted to the Day Type Number.

Name

Name of the day type

Field descriptions

day type

Unique identification of the day type

Name

Name of the day type

Shift time

Indication of start times and end times of the respective shift

Please note: For technical reasons a shift must not start or end at 0:00.

Type

At the moment this field is not relevant for processing.

Breaks

Indication of breaks (times of beginning and end) within the current shift.

If breaks are determined the times accrued in these periods of time are “cut out” of RPA time accounts.

Number of labor capacities

Production: This setting is only relevant in connection with the planning of group workplaces in the graphic planning board. The group workload in the shop floor scheduling module is calculated according to the defined shift calendar and the production capacities planned. Capacity 1 is accepted if no labor capacity is defined.

Setup: This setting is currently not taken into account.

Autom. status change to status 999 (only relevant for HYDRA-MDE configuration and only if HYDRA is applied)

This indicator can be used to automatically assign a machine status at the end of certain shifts (e.g. Friday evening) and in turn to cancel this status at the start of another shift (e.g. Monday morning).

No activation: Shift automatic will not be activated.

Activate after end of shift: The status 999 is automatically assigned to the machine at the end of the shift. If no status change occurs in the next shift, the status 999 remains set for the entire shift. Note: If the "Production" status is applied at the end of a shift, it remains set in the next shift. In this case, status 999 is not assigned.

Deactivate at beginning of shift: At the beginning of a shift the terminal automatically switches either to the status "not assigned" or to the status that was present before having activated status 999 – depending on whether the extended weekend automatic option is activated within the MDE configuration of the machine configuration. The possible status 999 will be finished.

The status 999, which indicates e.g. weekend, should be assigned to the resource performance account "free breaks" to prevent it from falsifying MDE evaluations.

14 Days Off

Overview

HYDRA menu	Master data → Production control → Days off
FEDRA menu	Detailed scheduling → Master data → Days off
Transaction code	wfmf
Function authorization	mdwfmf

Entries of dates for all days off of the corresponding year are written in the list of the days off. The list of the days off is used for the distinction between working days and free days when assigning day types to a year model. You use the list "Days off" to specify working days and days off when you assign day types in a year model.

Purpose

Once you have created the list of days off, it is available for all year models. Thus, the list has once to be created at the beginning of the year and then all year models of the respective year may access the list.

You must first create the list of days off before you create the year models. If entries are changed retroactively in the days off list these changes will only affect year models that are created after this modification. Year models that have already been established remain unchanged and, if necessary, the user has to adjust them manually.

Selection criteria

The application provides the following selection criteria:

Date from/to

Date restrictions

Field descriptions

Date

Date of the non-working day

Description

Description of the non-working day

Year

Year for which the day off was defined

15 Year Model

Overview

HYDRA menu	Master data → Production control → Year model
FEDRA menu	Detailed scheduling → Master data → Year model
Transaction code	ymmf
Function authorization	mdymmf

The year models are used as shift calendar of workplaces and machines.

Deactivating the calendar grid

Function authorization	ymmf_nocal
------------------------	------------

The display of the calendar grid in the application *Year model* provides a clear overview of the different day types used.

If a great number of year models is used, the calendar grid display can considerably slow down the display of the year models. In this case, assign the function authorization ymmf_nocal to the users. They can then deactivate the calendar grid display.

Purpose

You require a year model to assign a shift calendar to a machine/workplace. In the year model, a day type is assigned to each working day. The previously defined non-working days can be ignored (no shifts). Exactly one year model can be assigned to each machine.

Note the following

- If a day type is changed, this modification directly affects all machines assigned to this day type within the year model.
- Only change the year model for future days/periods. **We strongly advise against changing the year model for the current day.**

In the year model, a day type is specified for each day. The calculation of times in the system is based on this day type.

Requirements

Create the following master data before creating a year model:

- The days off must be specified.
- The day types must be specified.

Selection criteria

The application provides the following selection criteria:

Year from to

Narrows down the year models to the selected years.

Field descriptions

Year model, designation

This field shows the year model that is currently edited and its designation.

Factory calendar

You can set the identifier *Factory calendar* for one year model only (maximum). The factory calendar is used for the planning/[Scheduling](#). If you use the shop floor scheduling, you must therefore define a valid factory calendar for every year.

The year model, which is specified as factory calendar, does not affect the data collection.

Graphic display

The graphic display of the year model shows for every day of the year the relevant day type, which is assigned to this day. The background coloring is as follows:

Monday - Friday	gray
Saturday, Sunday	orange
Days off	red
Current day ("today")	dark red

The assigned day type is displayed for each day (number of day type). If a day does not have an entry, then no day type is assigned to this day. These days are treated by the system as without shifts.

Assignment

At the bottom, the dialog provides functions to assign day types to the days of the year model. You can use the *Weekdays* or a specified *Rhythm* to assign day types.

The section below describes how to assign day types to the days of a year model: See *Specification of a year model*.

Specification of a year model

Proceed as follows to specify a new year model:

Select the editing function *Insert*. The editing dialog opens. Select tab *Weekdays* or *Rhythm* and assign day types to the year model.

Weekdays

Date from ... to ...

A day type is assigned to the period of time specified here.



If you select *Weekdays*, the graphic display of the year model automatically selects all weekdays (Monday to Friday), which are included in the specified period.



You can also select by single weekdays. The graphic display of the year model then selects the days specified via selection for the period of time specified above.



If you select *Weekend*, the system automatically selects all Saturdays and Sundays of the period of time specified above.



You can also select either Saturday or Sunday only. The graphic display of the year model then selects the days specified via selection for the period of time specified above.

Include public holidays

If this option is enabled, all public holidays (defined as days off) are included in the selection of time.

Example:

The user selected *Weekdays* and *Include public holidays*. When the user then specifies a day type for the selected days, all public holidays on a weekday (Monday to Friday) are also assigned this day type.

Exclude public holidays

If this option is enabled, all public holidays (defined as days off) are excluded from the period of time specified.

Example:

Weekdays and *Exclude public holidays* are selected. When the user then specifies a day type for the selected days, all public holidays on a weekday (Monday to Friday) are not assigned this day type.

Public holidays only

If this option is enabled, *only* public holidays (defined as days off) are selected in the specified period of time.

Example:

Weekdays and *Only public holidays* are selected. When the user then specifies a day type for the selected days, only the public holidays on a weekday (Monday to Friday) are assigned this day type. All other public holidays, which are at a weekend, and all normal weekdays are not assigned any day type.

day type

Enter the day type in this field that you want to assign to the days previously selected.



Click this button to assign the entered day type to the selected days. The selected day type is then displayed in the year model for the days selected before.



Click this button to delete a day type assignment.

Note

There is another method to select days or a period of time without using the selection fields. Directly select the first day in the graphic year model, press the left mouse button and keep it pressed while dragging the mouse pointer to the last day. Here, release the mouse button. All days between the first and the last day specified are then selected (all days in chronological order).

You can add further days to this selection, if you click specific days while holding down the Ctrl key.

Rhythm

Use this tab to define a rhythm that you want to assign to the year model in a specified period of time (from - to or the days selected in the graphic display).

Step by step, define the rhythm and specify the day type and the number of days (for example: day type 4 for 1 day, day type 1 for 3 days, day type 2 for 1 day, day type 0 (no day type) for 2 days).

Apply the defined rhythm to the year model.

Rhythm duration

After the specified number of days, the rhythm used to transfer the day type to the year model is repeated.

Date from - to

The defined rhythm is assigned in the period of time specified here.

Duration

The day type is transferred to the year model for the specified number of consecutive days.

day type

Enter the day type in this field that you want to assign.



Click this button to assign the previously defined rhythm to the selected days.

From <date from>, the <day type> is applied for the number of days specified in <duration>. The day type is applied from the first <duration> day and is repeated after <rhythm duration> days.

The <day type> is then displayed in the graphic display of the year model for the days selected before.

Then, the <date from> is moved by the days specified in <duration>. From the new <date from> onwards, you can then configure a new rhythm.

How to proceed in case of changes of shift calendars - Recommendations

It can be necessary to make changes of day types and/or year models for operational reasons. Examples:

- Reorganization of shifts (e.g. from 2-shift operation to 3-shift operation)
- Change of shift times

In general, if the day types and year models are configured or used improperly, this can lead to unwanted results – similar to all system configurations.

Therefore, we strongly advise against changing day types that are currently being used (i.e. changes affecting the current day).

In this case, we recommend to proceed as follows:

- In general, do *not* make changes on the very day when the changes become active.
- Call the application *Day types*.
- Create a *new* day type or change an existing day type, but not the active day type (i.e. the assigned day type). (We urgently advise against changing existing day types that are already assigned to one or several year models.)
- Call the application *Year model* and open the year model where you want to make changes.

- Assign the new day type. (Important: You must select a date in the *future* as the day when the day type first applies).

16 Order Types

Overview

HYDRA menu	Master data → Order → Order types
FEDRA menu	Detailed scheduling → Master data → Order types
Transaction code	ot
Function authorization	mdot

Purpose

You use this function to create or modify order types in the system.

Integration

You use order types to classify orders according to their use. For each order type, different control information is stored, which is important for the order management. You configure the possible order types in the system that control the system behavior. Set the order type in the order header and also store this information with the operations (for information purposes).

Assign a category to each order type. A category combines similar order types.



We strongly recommend to copy order types only within one category.

Selection criteria

The selection panel provides the following selection criteria:

Category

See field description

Active

This is a *tri-state* checkbox which enables you to select either only active order types, inactive order types or both.

Field descriptions

General tab

Order type

Defines the order type. The parameters of this configuration specify, among other things, system behavior.

Name

Name of the order type.

Active

If this checkbox is set, the order type is available and you can use it. The system verifies if this checkbox is set when you attempt to create or modify an order.

Category

The category is used to classify and to combine similar order types. The category is a logical umbrella term. Possible values:

PO	Production order Standard production orders, which are created for example with an interface in the system, are assigned to the category production order.
PJ	Project order Projects are characterized by their uniqueness. A project order combines the project steps that are carried out during a project.
PM	Maintenance order You use maintenance orders to plan and/or record maintenance activities.
KP	Capacity order Use orders of this order type/category for capacity reservations in the shop floor scheduling.
GK	Overhead cost order Use overhead cost orders to collect overheads.

Category icon

You can assign an icon or symbol to the order type. Various functions/evaluations/reports (e.g. order overview) display this icon.



The icon size must not exceed 16x16 pixels. The file must be in BMP format (16 colors).

Dialog control

Here, you can store a dialog control that deviates from how the system behaves by default. This dialog control can be referenced in the dynamic dialog. You can then show or hide input fields on the terminal as required.

**Note**

You can also store this setting for the workplace (machine/workplace configuration). When a terminal dialog is opened, the dialog control for both order type and workplace are integrated. For this reason, we recommend that dialog control be defined either for the order type or for the workplace.

Authorization for the editing of inventory data

Reserved; currently not used.

Authorization for the editing of collected data

Reserved; currently not used.

Planning tab**Planning**

This setting specifies whether and/or how orders of this type are relevant for detailed scheduling. The following settings are available:

N No planning

T Scheduling only

The order is scheduled. This means that the lead time is calculated using the process times and then the dates are set accordingly. The (lead time) scheduling does not integrate competitive situations.

F Scheduling and detailed planning

The order is relevant for detailed planning: scheduling processes and automatic and/or interactive planning processes integrate this order.

Note: The automatic planning process plans operations for workplaces. This process integrates the workplace's capacity and assignment status. Consequently, it might be the case that you cannot plan the OP on the planned date (resulting from lead time scheduling).

**Note**

Whether or which operations are in fact integrated during scheduling or detailed planning also depends on the configuration of the same name in the *processing code* or the *order-related status configuration*.

Planned dates

Specifies how planned dates behave during (re)scheduling:

O Planned dates are not overwritten (by default).

T Planned dates are overwritten by scheduling.

Setup time in shop floor planning

Subject to the settings configured for the order type, you can choose from the below-mentioned options to present the static setup time of interrupted operations in the graphic planning board (interactive planning):

- „ ” Target setup time
- R Remaining setup time: Target setup time – RPA07 (posted setup times)
- B In case the operation has already been started (i.e. it is now interrupted), a setup time of 0 is assumed.

Running operations do not take into account the setup time.

These settings only apply for interactive planning.

User field for requirements calculation formula

This field is only relevant in connection with the license for multiple assignments. Further information can be found in the relevant [documentation](#).

Scheduling without implicit order release

This setting prevents order release (setting the status using the control indicator "V") during the first scheduling run.

Consideration of production variants when transferring OP

This processing is only active if the license for production variants is available.

E The system identifies whether a valid production variant exists at the time the OP is transferred/created and the following data is transferred from the production variant:

- Target cycle if the target cycle transferred via interface is 0.
- Partitioning if the partitioning transferred via interface is 0.
- Tool if included in the production variant.

If the system cannot identify a valid production variant, then the operation is not created.



This functionality is only available if the following conditions are fulfilled:

- Interface consulting services are required in advance.
- A customization must have been ordered.
- Production resources and tools are not transferred to the system

For information, also refer to the document

[Considering of production variants when transferring OPs.pdf.](#)

- | | |
|---|--|
| P | The system verifies if the transferred data matches a valid production variant (validation check) at the time an OP is transferred/ created.

If the system cannot identify a valid production variant, then the operation is not created. |
| N | No identification/ validation check |

Considering production variants in planning

This processing is only active if the license for production variants is available.

- | | |
|---|---|
| E | The system identifies if a valid production variant is available in graphic planning at the time an operation is planned. The default values of the production variant selected are used. |
| P | The system checks if a valid production variant is available in the system when the operation is scheduled (validation check). You cannot select a production variant and the default values of the production variant are not transferred. |
| G | Group-specific verification. During scheduling, the system checks like with option "P" if a valid production variant is available in the system for the OP that is scheduled (validation check). And it is checked if the group stored for the OP is identical to the group of the production variant. This check ensures that an OP can only be scheduled in the originally specified group also in case a production variant is used. |
| N | Not used. |

This processing can be controlled in more detail on the level of capacity groups. For further information refer to the relevant documentation dealing with the configuration of [Groups](#).

Processing tab

Sequencing list

This setting specifies whether the sequencing list at the terminal shows operations of orders with this order type.

The following settings are available:

- | | |
|---|--|
| F | The list shows fixed operations. You can set the <i>fixed</i> indicator manually in the shop floor scheduling. |
| N | No; the sequencing list on the terminal generally does not show orders of this order type. This makes sense, for example, with overhead cost operations. |

**Notes**

With all options except option "N", the "sequencing list" configurations of the same name defined in the *processing code* and/or the *order-related status configuration* are always used additionally.

In general, the sequencing list does not show blocked OPs.

Recordable

This option specifies whether operations of orders with this order type can be recorded, i.e. posted.

- | | |
|---|--|
| Y | Yes, operations can be recorded. |
| | Whether an operation can really be recorded also depends on the setting with the same name in the processing code. |
| N | No; Operations may not be recorded. If you attempt to log on any such operation is rejected with a validation error. |

Combined logon of order

You can use this option for "overhead cost operations".

- | | |
|---|--|
| Y | This order type links the operation with the person. The posting behavior is identical to that of a group workplace (GWP). |
| T | No overhead cost processing. |

Log on OP with start of shift

This option specifies the logon behavior of operations when a shift starts.

- | | |
|---|--|
| Y | Logs on operation automatically when the shift starts. |
| N | Does not log on the operation when the shift starts. |
| X | Processing depends on the relevant setting in the workplace/ machine configuration |

**Note**

This processing is only relevant for "MDE workplaces". An MDE workplace must meet the following conditions:

- The workplace must be assigned to a terminal.
- The terminal must be configured in *MDE* operation mode.

(only applies if HYDRA is in use)

May OPs of the order be terminated?

This option defines whether OPs of this order type may be finished. The following settings are available:

- | | |
|---|--|
| Y | Yes, you may finish the OPs. |
| S | Checks the status of the preceding OP. In this case, the status of the preceding OP must be assigned the option "successor can be logged off". |
| N | No, operator shall not finish the OP. A validation check is run when you attempt to log off an OP. |

**Note**

It can be useful to configure overhead cost orders so that they cannot be logged off via the dialog *Log OP off*, since their operations should exist as cost collectors over a longer period of time. In this case, you can set this option to "N" to prevent the OP from being logged off.

Log person off when shift ends

Use this option to configure the personnel-related data collection at MDE workplaces. Since HYDRA MDE generates fully automatic shift terminations via the terminals, you can set whether the persons logged on at the workplace are to be logged off automatically at the end of the shift or remain logged on (only applies when using HYDRA).

- | | |
|---|--|
| Y | Always log off staff when the shift ends. |
| N | Always save staff when the shift ends except for manual logoff. |
| X | Processing depends on the relevant setting in the workplace/ machine configuration |

**Notes**

This processing is only relevant for "MDE workplaces" (see above).

Set this option to "N" if the combined logon of order (see above) is active.
(only applies if HYDRA is used)

Serial number obligation

This option defines whether an order requires a serial number, and if so, how the serial number should be entered.

- | | |
|-----------|--|
| N | No serial numbers required. |
| + (plus) | Serial numbers required, "positive" entry, i.e. the yield is entered with serial numbers. |
| - (minus) | Serial numbers required, "negative" entry, i.e. the scrap or problem quantities are entered with serial numbers. |

**Note**

The option "Serial number obligation" specifies at the processing code or the operation directly as of which operation you have to enter the serial number.

Assignment of serial numbers

This option is only relevant if the option "Serial number obligation" is not set to "N". The option defines how serial numbers are allocated.

- | | |
|---|--|
| P | The PPS system transfers the serial numbers. |
| H | The system generates serial numbers according to the number range configuration. |

**Note**

The option "H" is provided for future upgrades/modifications. Currently, the system only supports the option "P".

Order is not logged off automatically

Processing can take place subject to the termination of the order (header). You can use this option to prevent the order from being set to finished when the last recordable operation was finished.

- | | |
|---|--|
| N | Finishing all recordable OPs automatically terminates the order (setting the order status to the control indicator "E"). |
| Y | Does not log off order automatically, i.e. finishing all recordable OPs does <i>not</i> automatically terminate the order. |

**Notes**

Currently, the status of the order (header) only affects:

- the deletion of order data or
- the transfer of order data into long-term tables.

You must set this option for the GKP and GKM order types (order types for waiting period orders). Reason: waiting period operations cannot be *recorded*. If you changed the status of a waiting period operation or if you deleted a waiting period operation, the system would set the order status to the control indicator "E". Conversely, this means you have to set a waiting period order (header) explicitly to finished (MOC function "Change status").

Milestone processing only for the last operation

Activates milestone processing.

Show OP info when logging on

Defines whether or not a notification or additional information is shown when logging the OP on to a Windows terminal (CTWIN, AIP). (only applies if HYDRA is in use)

Parallel order posting

This option specifies if linked operations are logged on, interrupted or logged off when you log on, interrupt or log off an operation.

Linked operations must have the following properties:

- You must specify the link type: H = main OP; N = sub-OP
Please note: You cannot set this field via the MOC.
- You must maintain the number of the main operation:
 - Sub-OP: MES order number of the main OP
 - Main OP: MES order number of the main OP (references itself).

You can find the field "main operation" in the operation tab "CBM".

The linked operations are logged on to the same workplace where the first operation has been logged on. For this reason, the option "Logon of several OPs" must not be set to "N" in the [Workplace and resource configuration](#).



The function does not support operations that require batch management, inspection operations or operations with assigned resources.

The terms "main OP" and "mother OP" are used as synonyms.

Plausibility tab

Check specifications in backlog of orders

This option activates the following validation check when logging on operations:

- | | |
|---|---|
| Y | <p>You can only log on the OP to the planned workplace.</p> <p>If you check this option, then during logon the system checks the workplace number entered at the terminal and compares it to the workplace number defined in the system order backlog. You can only log on the OP to the planned workplace. If you want to produce an operation at a different workplace (e.g. in the event of a machine breakdown), you have to replan this operation in graphic shop floor scheduling or in order sequencing.</p> |
| G | <p>You can only log on an operation to the planned group, i.e. you can log on the operation either to</p> <ul style="list-style-type: none"> - the planned workplace, - a workplace belonging to the group or to - a workplace planned for the group (still in pool of groups). |

- K OP logon only permitted either on a planned workplace or on a workplace of the same category. **Please note:** This check requires that a category is defined for both workplaces (field "category" must not be empty), i.e. for the workplace where you planned to log on the operation AND for the workplace where you actually logged on the operation.
- S Group control: This option differentiates between the following initial situations:
- Situation 1:
- You have planned the operation for a workplace. You can log on this operation either to the planned workplace or to a workplace of the same category.
- Situation 2:
- You have planned the operation for a group. Consequently, the operation is still in the pool for the group (pool of groups). You can only log on this operation to a workplace that has the same category as the first available (not locked) workplace of the planned group.
- N No validation check

Check status of preceding OP

This option activates a check that specifies whether or when the current operation may be logged on.

Minimum send-ahead quantity of preceding OP

This option defines whether a validation check for the minimum send-ahead quantity should be run when logging on the operation.

- M Check the minimum send-ahead quantity when logging on operations. In this case, a minimum send-ahead quantity (yield) must have been posted to the preceding operation so that the succeeding operation can be logged on. The minimum send-ahead quantity is defined in the preceding operation.
- N No check is run.



You have to enable the relevant configuration in the order type in order to verify the minimum send-ahead quantity when logging on OPs. Configure the processing code accordingly to plan overlapping operations based on the minimum send-ahead quantity (or the lead time). You can enable this function in the processing code while customizing the system.

When the system checks the minimum send-ahead quantity, only the **yield quantity (primary quantity unit)** collected so far is used.

Quantities are not converted. For this reason, make sure that adjacent

operations have the same primary quantity unit.

Example:

Operation 0100	Target quantity 1000	Send-ahead quantity 50
Operation 0200	Target quantity 1000	

If checking is enabled, operation 0200 can only be logged on as soon as a yield (in primary quantity unit) of at least 50 has been uploaded/confirmed for operation 0100.



The system does not check the operation status of the preceding operation. You cannot log on the current operation, in case the preceding operation has already been finished, but the send-ahead quantity has not yet been reached.

Only OPs of the same order type can be logged on simultaneously

If this option is set, you can only log on operations in parallel that are assigned to the same order type.

The system will run this check if this option is set at least for one operation, i.e. the operation to be logged on or the operations that have already been logged on.

Check if shop floor papers are printed

If you set this option, you enable a validation check when logging on operations verifying the print status of time tickets. Under proper conditions, this function can help prevent erroneous postings or rather "guide" production.

- | | |
|---|---|
| Y | You can only log on an OP if its shop floor papers have been printed. |
| N | No validation check |



Note

A corresponding identifier at the operation indicates the print status. It can be set when printing the time tickets (additional license).

Check against M/O relation when person logs in

You can define the *maximum number of employees needed to produce this operation (M/O relation, production)* in the system's operation data.

The system verifies the following when staff logs in:

- | | |
|---|--|
| Y | Machine/ operator relation. When a person logs on to an operation, a check is run to determine if the maximum number of employees has already been reached. If this is the case, the system rejects the login. |
| N | No check. |

K Planned; currently not used.



Note:

The check is not performed for group workplaces.

If the M/O ratio defined for the operation is not an integer, the value is rounded up to the next whole number.

Checking also includes advance logons of staff.

Quantity check of send-ahead quantity

G If this identifier is set, then the system runs a check for manual quantity postings at the terminal to determine whether the sum total of the previously entered *yield* and the currently entered yield is greater than the already produced yield of the preceding OP. If this is the case, the system does not allow the posting with the entered yield quantity.

S If this identifier is set, then the system runs a check for manual quantity postings at the terminal to determine whether the sum total of the previously entered *total quantity* (yield + scrap + rework + open quantity) and the currently entered quantity (yield + scrap + rework + open quantity) is greater than the already produced yield of the preceding OP. If this is the case, the system does not allow the posting with the entered quantity.

N No check.



Note

This requires that the current operation and the preceding operation each have the same primary quantity unit.

Quality tab

CAQ area

This field is currently not used.

Generate CAQ inspection steps

This field is currently not used.

Confirmation tab

Upload order postings

Upload

This identifier specifies whether entered data (postings) of orders pertaining to this type must be uploaded to the ERP system or not.

- | | |
|---|---|
| Y | Uploads postings. Use the following options to define which postings are uploaded/confirmed and/or how they are uploaded. |
| N | Does not upload postings. In this case, the following options are irrelevant. |

OP logins

If you set this option, operation logins (record type "A" postings) are uploaded to the ERP system. The same structure is maintained during an upload as for postings, but in this case the quantity and duration fields are set to 0.

Postings of type "A" records are generated for each operation login, including those automatic logins at the time of automatic shift change processes.

Please note: If the PP-PDC/KK3/KK4 interface uploads the postings to SAP, the OP logins are assigned the SAP record type L10.



As of program version myerprck.exe/out V8.1.1.103, you can use SAP-compliant record types to upload operation logins to SAP. You can find details in the following documents:

- [HYDRA setup configurations for PP-PDC](#)
- [HYDRA setup configurations for KK3](#)
- [HYDRA setup configurations for KK4](#)

Postings of part quantities (partial confirmations)

If this identifier is set, part quantities (record type "T" postings) are uploaded/confirmed to the ERP/PPS system. In this case, the posting generates U/E records that do not contain any quantities. Only the T-records still contain quantities.

As far as the ERP/PPS system is concerned, uploads based on T-records are represented at the upload interface in the same manner as record type "U" posting records. However, they only contain quantities, no durations. Durations continue to be uploaded from the postings of record types "U" and/or "E".

If the identifier is not set, the U/E records generated by the posting process are, in addition to the durations, also given the accumulated quantities of all T records generated between login and interruption/ logoff.

Upload of scrap including reason

Unlike with the "partial confirmations", the "Upload of scrap including reasons" only uploads the scrap including scrap reason. The record type U and/or E postings confirm the yield, rework quantity and problem quantity.

Use this option to upload/confirm the recorded scrap quantities and scrap reasons based on the recorded T records to the ERP system. The system only uploads yield, once the operation has been interrupted and logged off.

Please note: If both options are set, the "Partial confirmations" option has the higher priority.

Approved order postings only

If this option is set, order postings are not uploaded until they are approved. After 5 days at the latest (standard setting, [can be changed via the parameter /ABZEICH= in the program myerprck.exe/.out.](#)), non-authorized postings are also confirmed/uploaded.

Upload company of actual workplace

If you set this option, the system uploads/confirms the company that is defined in the machine master record of the actual workplace, instead of the company that is defined for the workplace in the orders/operations table (auftrags_bestand).

Order postings instead of batch postings

For MPL machines, batch postings are uploaded by default (record type "H" postings). Instead of batch postings, this option allows you to upload order postings (record type "U"/"E" postings).

Upload type

Reserved

Change after upload

This option specifies how the changes/corrections are processed that are made to postings, which have been uploaded to the ERP/PPS system. This option is evaluated when you use the editing functions for postings and for events.

- | | |
|---|---|
| Y | Changes/ corrections to already uploaded postings are allowed; changes are also uploaded to the PPS system. |
| E | Changes/ corrections to already uploaded postings are allowed; however, these corrections are not uploaded to the PPS system. |
| N | No more changes/ corrections are allowed to already uploaded postings |



Note:

If you edit a posting, the system evaluates this option with direct reference to the relevant operation. But if you edit an event, the system uses the last operation with "N" or "E" and locks all previous events irrespective of the order type.

Do not set this option if you use the SAP PP-PDC interface because this interface does not support cancellations/corrections. Please contact MPDV to discuss options available to nonetheless upload corrections to SAP.

Cancellation message job end first

Currently not used.

Upload of waiting period postings

If this option is set, then postings identified as "waiting period" (ade_protokoll.karenz = 'P' or 'M') are uploaded to the PPS system.

If you only want approved waiting period postings to be uploaded, then you must set the option "Upload approved order postings only" or "Upload approved personnel postings only".

Upload of batch logon postings

Please note: This option is released for uploads via the HYDRA HY72PPS interface only. You must not use this option with the interfaces PP-PDC, KK3, KK4 or HY71PPS!

This option uploads batch logons via the HY72PPS upload interface. The upload record does not contain quantities/ durations and the logon/ logoff times correspond to the posting time.



Please note: In HYDRA standard processing, order logons and batch logons are uploaded first, i.e. if an order is logged on twice and interrupted once between two posting intervals, then the two logons are uploaded before the interruption.

Personnel postings

Personnel postings

If this option is set, then record type "B" postings (personnel postings) are uploaded to the PPS system.



Note:

This option may not be set for customers who use the SAP PP-PDC interface.

Upload personnel postings only.

Upload personnel postings only.



Note:

If this option is set, then record type "B" and "E" postings are uploaded to the PPS system.

Approved personnel postings only

If this option is set, record type "B" postings (personnel postings and personnel waiting period postings, if configured for uploading) are not uploaded until after approval. After 5 days at the latest (standard setting, [can be changed via the parameter /ABZEICH= in the program myerprck.exe/out.](#)), non-authorized postings are also confirmed/uploaded.

Confirmation of planning changes in shop floor scheduling

Upload PPS operations only

Upload PPS operations only

If you activate this option, the operations created in HYDRA are not uploaded.

Upload delayed operations only

Upload delayed operations only

If you activate this option, only delayed operations are uploaded.

Upload operations with deviating assignment only

Upload operations with deviating assignment only

If you activate this option, only operations are uploaded that have been planned for another workplace than the workplace initially transferred.

You can use the option *Upload PPS operations only* no matter whether the other options are enabled or not. But you can only enable one of the two options *Upload delayed operations only* and *Upload operations with deviating assignment only*.

Quality tab

CAQ area

This field is reserved for future functions.

Generate CAQ inspection orders

This field is reserved for future functions.

Options tab

Priority control

Y	The priority management integrates orders of this order type (see configuration Order group). This means that if you create a new order or manually change an order in HYDRA, the system runs a check to identify whether the assigned priority is allowed. If the maximum number has been exceeded, the action is rejected. The system only runs the priority check when changing orders/operations if you actually changed the priority. If the order is transferred from the ERP interface and the maximum number is exceeded, the order is not be refused as a result of this validation check. In this case, however, the priority is automatically set to 1.
---	---

Notes

This check is only performed if the option *Priorities* is set to 'U'.

The check only uses the orders where the option *Priority check* is enabled for the order status in the [Configuration of the order status](#).

N Priority management does not integrate orders of this order type.

Priorities

U The priority is controlled via the order. This means that a priority set for the order (header) is applied to all operations of the order.

G The priority is controlled via the operation. This means that a change of priority for the order (header) does not change the priority of the operations of this order.

Order postings need to be signed off

This option specifies that order postings for operations with this order type must be signed/approved (see also the manual of escalation "ANR.UNCERTIFIED_BOOKINGS"). If the postings are not signed/approved, the foreman's checklist includes the corresponding log records. This option does not affect uploads via interface.

Personnel postings need to be signed

This option specifies that personnel postings for operations with this order type must be signed/approved (also see manual of escalation "ANR.UNCERTIFIED_BOOKINGS"). If the postings are not signed/approved, the foreman's checklist includes the corresponding log records. This option does not affect uploads via interface.

Order type for composition

The *composition* takes into account charging orders. Charging orders are assigned to melting orders. In this case, the *planned order number* field of the charging order includes the order number of the melting order.

"C" = Charging order,

"M" = Melting order

Tab User fields

The system can be customized to enable user fields for the object type "AUART".

17 Order Status Texts

Overview

HYDRA menu	Master data → Order → Order status texts
FEDRA menu	Detailed Scheduling → Master data → Order status texts
Transaction code	osst
Function authorization	mdostt

You can use the application *Order status texts* to configure the description of the relevant order status.

Purpose

In the status text dialog, you assign a descriptive text to all possible order statuses. These descriptions are then used in the [order status assignment](#) performed later. The objective is to use standard status texts for all order types.

Procedure

Create the status texts that you would like to use for status assignment later.

Assign a unique status text number and an informative text.

Then, categorize each of the status texts to an order or operation status.

Integration

The texts defined here are displayed on the client based on the order status assignment performed in various evaluations/reports.

Selection criteria

The application provides the following selection criteria:

Status text designation

Search for a status text by status text designation. You can also run a search using wildcards.

Field description

Field descriptions status text

Status text identification

Designation

Status text

18 Order Status Assignment

Overview

HYDRA menu	Master data → Order → Order status assignment
FEDRA menu	Detailed Scheduling → Master data → Order status assignment
Transaction code	ost
Function authorization	mdost

Purpose

You can use the *Order status assignment* to configure the order statuses for the different order types. The order status provides the current status of the order.

Integration

Because order data are recorded in the system based on the operation, the system manages one status for each individual operation. The status indicates whether the operation has not yet begun, for example, or whether it has begun, was interrupted or has already been finished.

In addition, a status is also kept in the order header, which provides information on whether the order has not yet begun, whether it is currently being processed or whether it is already finished. At the time the last operation that can be posted is finished, the status is set to finished in the order header. Operations that cannot be posted will not be automatically finished when this occurs.

Both order and operation statuses can be configured. To do so, the status texts must be defined first. Then, the status texts are assigned to the individual statuses. What must be kept in mind is that for each order type, the order status or the operation status must be configured with one control indicator each (exception: statuses with a control indicator "S" may exist multiple times).

All possible order and operation statuses are configured in status assignment and are assigned to status texts using unique status numbers. The selection dialog allows you to select and display statuses that have already been assigned to an order type.

Requirement



The order status texts and order types must be created first.

Selection criteria

Status type

Selection of the assigned status type

Status text

Defined status text

Order type

Order type that was assigned to the status

Field descriptions

This dialog is used to define the statuses used in both the order (header) as well as in the separate operations. There is also another option that allows you to define an operation in relation to the secondary status. To differentiate which identifier is relevant for which status type, the corresponding status type is written in parenthesis after the field:

- A = Order status
- G = Operation status
- S = Secondary status

Order type [A G S]

The status is configured based on the order type. This means that for each order type the relevant order and operation statuses must be defined. Enter a valid order type in this field, for which the status should apply.

Status type (A/G/S)

A	Status refers to the (entire) order
G	Status refers to the operation
S	This is the preparation status (=secondary status) at the operation.

Status (A/G/S)

This is the externally valid status.

Status tab

Status text no. (A, G, S)

Reference to the order status text table

Symbol (G)

Symbol assigned to the status. It is displayed in different functions/ evaluations (e.g. order overview).

The following values are allowed (case sensitive):

Color of the LED	Value that needs to be entered in the editing dialog.	Value shown in the detail panel.
Light green	l.bmp	LED_LIGHT_GREEN
Dark green	e.bmp	LED_GREEN
Gray	v.bmp	LED_GREY
Yellow	u.bmp	LED_YELLOW
Yellow/ light green	f.bmp	LED_YELLOW_GREEN
Pink	n.bmp	LED_PINK
Black	a.bmp	LED_BLACK
Blue	p.bmp	LED_BLUE
Red	x.bmp	LED_RED

Color (G)

Color of the relevant status (text). At this time, this color is only used in the shop floor scheduling.

Entry tab

Control

Status type A (order status)

V	Prepared Release (for automatic release)
L	Running At least one OP was started
E	Finished All OPs are finished or deleted.
S	None All others

Status type G (OP status)

V	Prepared
---	----------

"Release indicator" for status; before OP is edited for the first time

L	Running Is set automatically once OP is logged on
U	Interrupted Is set automatically once OP is interrupted
F	Autom. interrupted Is set by the server if you want the OP to be interrupted automatically at the time of a shift change (when using HYDRA and the MDE function on the terminal)
E	Finished Is set automatically once OP is finished
S	None Other status; not automatically set. Means that no work has yet been done on the OP

Please note

An OP has not yet begun if its status is set with a control indicator: "V" or "S".

Explicitly setting an order to the status using the control indicator "E" will result in all OPs being set to the status with the control indicator "E".

The control indicator "A" means "archived"; it is not set via configuration, but is instead a fixed setting made at the time the data are transferred into the long-term table.

Can be logged on

Check to assure if the CURRENT status has set this indicator. If the operation is interrupted/ finished, the field is ignored. .

Please note: If an OP or an order have been set with the flag "locked", then the OP may not be logged on under any circumstances (irrespective of this setting)

Sequencing list

If this status is set, the operation is shown in the sequencing list at the terminal.

Please note: if an OP or an order has been set with the flag "locked", then the OP may not be displayed under any circumstances (irrespective of this setting).

Successor can be logged on

If the indicator "Check of preceding OP" ("Plausibilities" index tab) is set to "S" = status at the order type, then this indicator is used to check whether the subsequent OP can be logged on.

The indicator relates to the preceding OP!

If, for example, an OP logon of the subsequent OP is not to be allowed until the preceding OP has at least begun, then this indicator must be set if the status is "running", "interrupted", "finished" (possibly "deleted").

If, for example, an OP logon of the subsequent OP is not to be allowed until the preceding OP has finished, then this indicator needs only to be set if the status is "finished" (possibly "deleted").

All preceding OPs resulting from relationships in ADE_AUFTRAGSNETZ, are checked in this regard.

Successor can be logged off

If the indicator "Can be finished" ("Entry" index tab) is set to "S" = status at the order type, then this indicator is used to check whether the subsequent OP can be logged off.

The indicator relates to the preceding OP!

Example 1: If an OP logoff of the subsequent OP is not to be allowed until the preceding OP has also *at least begun*, then this indicator must be set to "J" (y – yes) if the status is "running", "interrupted" or "finished".

Example 2: If an OP logoff of the subsequent OP is not to be allowed until the preceding OP has *finished*, then this indicator needs only to be set if the status is "finished" .

All preceding OPs resulting from relationships in ADE_AUFTRAGSNETZ, are checked in this regard.

Release of subsequent OP



The release is only processed if the operation status has changed after an operation posting (logon, interruption, logoff).

If this indicator is set, then setting the status will result in the status of the subsequent OP(s) being set to the release status, i.e. set to the status assigned with the control indicator "V".

This only happens if the subsequent OP(s) has (have) not yet been run, was (were) interrupted or has (have) been finished.

Thus, this option only makes sense if new operations are not created/ transferred using the status with the control indicator "V".

Planning tab

Planning

N	No planning (no transfer to the planning component)
T	Scheduling only
F	Scheduling and detailed planning (dispatching) - also includes the simulation

**Please note:**

The indicator "N" at the order overrides the indicator at the operations. Conversely, the indicators at the operations override the indicator "J" (y – yes) at the order

Posting tab**Posting order duration****Posting**

M Posting to the RPA of the workplace status
A posting to the RPA of the operation is made using the RPA of the status currently active at the workplace (default).

Posting RPA

Reserved; currently not used

Options tab**Initial status**

This relates to the initial status that should be set at the time an order (A) or an OP (G, S) is set.

Exactly one status must be set using this indicator for both the order as well as the operation.

Please note: If the initial status is not the status with the control indicator "V", then the status of the first operation as well as that of the order is set using the control indicator "V" after the first scheduling, provided that the indicator "Scheduling without implicit order release" is *not* set at the order type.

Status can be set manually

Reserved; currently not used.

Authorization

Reserved; currently not used

Change of secondary status

Change of secondary status allowed (J (y – yes)/N)

Priority check

Checking the priority in the priority management (order group)

If the priority control is activated in the configuration of the [Order types](#), then the system performs a check using the priority management ([Order group](#)) when an order is created or changed.

The check includes all orders with the status where this option is enabled.

Note: The priority is only checked if actually the priority was changed.

Alterable order data

If this status is set, the order or the OP may either be altered by the PPS system or manually or it may no longer be altered.

J = console and MLE interface (PPS=J) alterable

N = console and MLE interface not alterable

K = Only console alterable (not via the MLE interface)

M = Only MLE interface alterable (not via the console)

This option does not refer to the priority, notes, long texts, secondary statuses.

Deletable

May the order or the operation be deleted in this status (manually, via PPS system)?

Order

Yes, BUT only if the operations can be deleted, i.e. at no operation may a status be set that has a control indicator "L" AND for which the "deletable" indicator is set to "N".

Operation

If the OP is running, then the deletion flag is set.

Please note: Operations with a current status with a control indicator "L" or "F" may as a rule not be deleted, irrespective of this setting.

Action

Behavior indicating how the order or the operation should respond during the deletion process, i.e. at the time of deletion (now) or when a deletion flag is triggered:

Order

- | | |
|---|--|
| L | If something is physically deleted and all operations are physically deleted as well (order header controls the behavior) |
| E | Order is set to the status with the control indicator "E".
WARNING: Only the order header, i.e. for the OPs the indicator must also be set! |
| D | Order is set to status with control indicator "D" and is deleted/ archived using the archiving/ deletion program (data management) after the set period. How the operations are treated is described via the indicator RESET ACTION (LOESCHAKTION) at the status of each OP. |
| X | If the action was triggered by the MLE interface:
Order is set to the status with the control indicator "E".
WARNING: Only the order header, i.e. for the OPs the indicator must also be |

set!

If the action was NOT triggered by the MLE interface:

If something is physically deleted and all operations are physically deleted as well (order header controls the behavior)

Operation

- L Is physically deleted.
- D Is set to the status with the control indicator "D".
- E OP is set to the status with the control indicator "E" (fixed).
- X If the action was triggered by the MLE interface:
OP is set to the status with the control indicator "E" (fixed).
If the action was NOT triggered by the MLE interface:
OP is physically deleted.



The reset/deletion action should be the same for both the order header as well as for the OPs, because currently the settings each relate to the order or to OPs and the OPs do not "inherit" anything from their order header.

To assure that an order or an operation can no longer be deleted as of a certain status, the indicator must be set at all corresponding statuses.

19 Order Postings

Overview

Menu	Data collection → Corrections → Order-related postings
Transaction code	oboo
Function authorization	oboo

Available user fields		
Where?	Object type/user field key	Source (type)
Table and detail view	ADEPRO/SYSTEM	BDE log record (MF-D)
How to configure user fields?		Which user field types are available?

Purpose

The application *Order-related postings* is a very useful tool for any foreman, team leader, supervisor or other responsible person who wants to check/edit postings that have been recorded during production.

Integration

The order-related postings integrate all data that has been collected during production. The postings and results from production are listed here in great detail and can be reviewed and edited.

The list in the application shows all postings that are included in the selection made in the selection panel. The system only displays the postings of workplaces/machines (record types unequal "B") and of persons (record type "B") that are included in the user's responsibility area.



In case of staff postings (record type "B"), the system only checks the current responsibility area of the person and not the responsibility area that was valid at the point in time of the posting (the different versions of the HR master data are disregarded).

Requirements

Postings (logon, interruption, posting of part quantities/partial confirmation, logoff) must be collected for each operation.

The HYDRA basic setting *Maintenance on the basis of events* specifies which editing functions are available to modify postings.

If this option is *not* set, all postings may be edited using the editing function for postings, i.e. postings recorded manually and postings generated automatically by HYDRA due to the recorded events.

If this option is enabled, postings generated automatically in HYDRA (due to events) cannot be corrected nor deleted using the editing function for postings. Such postings can only be edited using the function *Event maintenance*. Using the editing function for postings, you can only correct or delete manually recorded postings.

Automatically generated postings and manually recorded postings

In HYDRA, there are automatically generated postings and manually recorded postings. Automatically generated postings are postings generated automatically by HYDRA based on the recorded posting events. In contrast to this, manually recorded postings are postings that are recorded manually using the editing function. They do not relate to events.



HYDRA does not generate or correct subsequent events for manually created or edited postings.

If you change/generate/delete a BDE posting (shop floor data collection), this does not have an impact on MDE postings (machine data collection)!

If you change/generate/delete a posting, this does not result in a recalculation of the actual cycle that relates to an operation.

Selection criteria

The application provides the following selection criteria:

Order

All postings for this order are shown when the order number is entered.

Operation

This selection criterion refers to the operation number. All postings are shown that are posted to the operation number that was entered.

Workplace

This selection criterion specifies the workplace number where the posting was made. All postings are displayed that were generated as a result of the events at this machine/this workplace.

Cost center

If you select a cost center, all postings are shown that were posted at the workplaces/machines of this cost center in the specified period of time.

Record type

It is possible to limit the selection by record type ("U/ E/ H/ B/ T records").

If no explicit selection is made, the system filters the records of type "U" and "E" for operations that do not require batch management. For operations that require batch management, the system only displays postings of record type "H", if no explicit selection of record types is made.

Person from ... to ...

If you enter a personnel number, all staff postings ("B records") can be displayed that were generated via personnel postings of the specified person. Also the manual postings, which were generated with the staff badge number, are displayed.

Order type

Select one or more order types to display all postings for orders of the selected order type.

Category

Select one or more categories (order categories) to display all postings for orders of the selected category.

Interruption reason

If you enter an interruption reason, all postings are displayed that were canceled or interrupted with this interruption reason.

Premium group

In HYDRA Incentive Wage LLE, you assign workplaces/machines or persons to premium groups. If you enter a premium group, all postings are shown that were made for persons or workplaces/machines of this premium group in the period of time entered.

Modified by/modified on

If you enter a person in the "modified by" field and an editing date, you can view all postings that a person edited on this day.

Incl. cancellation log records

This option is used to select and display cancellation log records (postings of record type "S").

Incl. original log records

Use this option to additionally display original postings (postings of record type "O").

Non-authorized postings

Use this check box to restrict the display to order-related postings that have not yet been authorized (record type U, E, H, T, B). This makes sense if postings should not be uploaded until they have been authorized.

Date from ... to ...

All postings made in the specified period are displayed.

The selection of a shift might issue a different result than the selection of a time for a specified date.



Note: This application only selects and edits data that is included in the online data area.

Shift

Enter a shift to select postings that were posted during a specific shift (according to the BDE shift model).

Note: The logoff time stamp of a BDE log record identifies the shift (shift date, shift number) the log record is assigned to.

Time

Use the time to further restrict the selection of postings. This selection criterion references the posting's start time. Please note in this context that this start date may definitely go back farther into the past for workplaces where the shift automatic function has not been activated.

If you enter several selection criteria, the data is displayed that matches all selection criteria.



Note on merged operations: If you have merged operations via MOC, ONLY the separate single operations can be displayed and changed. If you change/correct the log records of single workplaces, the status of the merged operation is updated.



Note on split operations: In case of split operations, the log records of the split operations are displayed. No log records are generated from the split operations for the split master. When log records of split operations are changed/corrected, the status of the split master is updated.

Toolbar



Authorize (function authorization: oboo.sign)

Function to [Authorize postings](#)



Machine-related postings (function authorization: mboo)

Click this button to call the application [Machine-related postings](#)

When you call the function, workplace, cost center, date/time and shift are transferred from the selection panel.



Order information (function authorization: orin)

Click this button to call the [Order information](#).



Inspection points (function authorization ipp)

Click this button to call the application [Inspection points](#)

Field description

In the table view, the posting's data is displayed including other additional information. In the editing dialog, only the data relevant to the posting is displayed.

Tab General

Record type

Record type of an order-related posting:

U order interruption: The interruption of the operation triggered the posting

E Posting of finished order: The posting is triggered, when the operation is finished.

B Staff posting: Personnel-related posting

H Change of batches: Posting that relates to batches. If you use the Material and Production Logistics (MPL), the postings for operations that require batch management always relate to the specific batch. In this case, the postings of record type U or E are not displayed if the record type is not selected explicitly.

T Partial confirmation: Posting of a part quantity (scrap with reason)

Order

Order number

Sequence

If sequences are used, the sequence is displayed here.

OP

The specified order/operation must exist; the status is not checked.

Split

If splits are used, the split number is displayed here.

Workplace

Number of the workplace/machine where the operation was logged on.

Cost center

Cost center of the workplace where the operation was logged on/posted.

Person

Personnel number of the person who triggered the posting.

This number must exist in HYDRA. In case of staff postings (record type "B"), it is mandatory to enter a personnel number. The field can remain empty with the other record types.

Employee group (only displayed in tabular view)

Assigned employee group according to the HR master data

Interruption reason

Status of the machine/workplace at the time of the interruption or completion posting. This relates to the workplace/machine status.

Internal MES batch number (only relevant when using MPL)

Produced batch number or entered batch number

Quantity index tab

Primary quantity

Yield

Yield produced in the primary quantity unit

Scrap

Scrap produced in the primary quantity unit

Scrap reason

Scrap reason

Rework

Rework quantity in primary quantity unit

Open quantity

Recorded open quantity in the primary quantity unit

Unit

Quantity unit, is taken over from the operation.

Note: The field cannot be changed in editing mode

Total cycles (only displayed in tabular view)

Cycles posted while the operation was logged on (only for postings of record type U, E)

Cycles during MUT (only displayed in tabular view)

Cycles posted while the operation was logged on and the status "Production" was available (only for postings of record type U, E).

Secondary quantity/tertiary quantity/basic quantity

Yield, scrap, rework, open quantity, unit

Posted quantities in the respective quantity unit.

Note: You cannot change the Unit fields in editing mode.

Durations index tab

Login

Date and time of the start posting that issued this posting.

Logoff

Date and time of the start posting or the end posting that issued this posting.



If you change the points in time *Login* or *Logoff*, then this does not change the saved points in time of "First logon", "Last interruption", "Last logoff" or "Status since" in the order or operation status.

Shift, shift date, shift start, end of shift

Number of the shift, shift date assigned to the posting according to the BDE shift model, and shift times.

Shift model (only displayed in tabular view)

Day type that was used for the (automatic) posting.

Usually, the day type of the shift model of the workplace is used here. If the basic setting Synchronize labor times with the person's BDE shift model is enabled, the day type of the BDE shift model that is assigned to the person is used here.

Shift type (only displayed in tabular view)

Shift type that was defined in the day type configuration.

Order-related RPA

Order-related times that were posted to the resource performance accounts.

Personal RPA

Personal times that were posted to the resource performance accounts.

Duration (only displayed in tabular view)

The posting duration is the sum total of the order-related times that were posted to RPA 1-11.

Sum (without breaks) (in the editing dialog only)

The value corresponds to the duration of the posting (column "duration" in the tabular view) and is calculated automatically from the sum total of RPA 1 to RPA 11.

Difference (in the editing dialog only)

The difference between the original duration and

- the new duration from the total of RPA1-11 or
- from the different posting times

is shown for information purposes.

Labor utilization

Time of labor utilization. The calculation depends on the option RPAs to calculate labor times in the [Basic settings](#).

Compensation with RPA

If you edit the durations and as a result you have a difference, this difference is displayed in the respective field in the editing dialog. To offset this difference, an RPA is specified. When you then save the changes, the difference is assigned to the specified RPA.

If you click the icon  in the *title bar* of the editing dialog, the difference is offset against the specified RPA.

Offset labor utilization

You can use the option *Offset labor utilization* to offset the difference against labor utilization.

If you have clicked the icon  and the personal RPAs have been offset, the RPA durations are totaled and the sum is entered in the field labor utilization.

In the [Basic settings](#), tab *BDE*, section *RPAs to calculate labor times*: If the option *Processing active* is enabled, only the RPAs are integrated that are enabled in this section of the basic settings when the RPA durations are totaled up.

In general, order-related resource performance accounts and personal resource performance accounts are displayed and can be edited separately.

In case of staff postings (record type "B"), the display of order and staff postings depends on the option *Proportionate RPA posting in personnel postings* in the [Basic settings](#).

- If the option *Proportionate RPA posting in personnel postings* is enabled in the basic settings, the order-related and the personal resource performance accounts are displayed for staff postings (record type "B").
- If the option *Proportionate RPA posting in personnel postings* is not enabled in the basic settings, only the personal resource performance accounts are displayed for staff postings (record type "B"). The order-related resource performance accounts are automatically set to the values of the person-related resource performance accounts.

Field descriptions in tab *Wage data*

te

Default individual time for labor time in [h/1000]. Is taken over from the operation (requires the relevant LLE license).

tr

Target specification for setup time in [h]. Is taken over from the operation (requires the relevant LLE license).

teb

Default individual time for the (machine) occupancy time in [h/1000]. Is taken over from the operation (requires the relevant LLE license).

trb

Default individual time for the (machine) setup time in [h]. Is taken over from the operation (requires the relevant LLE license).

Wage type

The wage type is taken over from the operation (requires the relevant LLE license).

Note: Not this value is uploaded as wage type at the ERP upload interface, but the value defined in the order backlog of the operation.

Premium group •

In HYDRA Incentive Wage LLE, either workplaces/machines or persons are assigned to premium groups (requires the relevant LLE license).

In this case, the specified premium group must be valid.

Operator position/function

Operator position entered for the posting (only relevant for personnel postings).

Premium indicator

Premium indicator entered for the posting (only relevant for personnel postings).

Settlement date

The system uses the settlement date to assign a posting to a settlement day for the calculation of the incentive pay and the PZE-BDE comparison function. If the settlement date is filled, the order-related posting is assigned to this day. If the field *Settlement date* is empty, the field *Shift date* is used for the assignment. If the shift date regularly does not correspond to the PZE assignment, the BDE shift models need to be adjusted.

The field is filled in as a result of the PZE work day evaluation or via the editing function for postings. This means that normally the field is initially empty for all postings of the current day and that it will not be updated until the next morning after the PZE work day evaluation has been run. As a result, postings for the current day may be incorrectly assigned to the previous day. The incorrect assignment will normally disappear on the next day.

The following rules apply for the assignment of the settlement date:

Order-related staff postings (B records)

Staff postings are assigned to the PZE personal day if the postings are within a time frame of +/- 2 hours to the rounded working times of the reporting person.

PZE	IN		OUT	
	21:47	22:00	6:00	6:17
	----- ----->t			
	IN ROUNDED		OUT ROUNDED	
Rounded working time	[=====]			
	-2	+2	-2	+2
B records	---[=====]---		---[=====]---	

If a staff posting lasts several days, it is assigned to the first PZE personal day, because usually a person has forgotten to log off (P_AB).

Order-related postings (U/E/T records)

Order-related postings are assigned to the PZE personal day of the reporting person if the end time of the postings is within a time frame of +/- 2 hours to the rounded working times of the reporting person.

PZE	IN	21:47	22:00		OUT	6:00	6:17
				----- ----- ----->t			
			IN ROUNDED		OUT ROUNDED		
Rounded working time			[=====]		[=====]		
		-2 +2		-2 +2		-2 +2	
		--- ---		--- ---		--- ---	
U/E records		[=====]		[=====]		[=]	

If a posting lasts several days, it is assigned to the last PZE personal day, because the reporting person is identified using the logoff and therefore the person logging off posts the quantities. If no personnel number is entered in the posting, the settlement day cannot be assigned.

User fields index tab

See [above](#).

Administration index tab

Modified by

Person who last edited the posting.

Modified on

Date when the posting was last edited.

Name

Name of the person who last edited the posting (according to the user administration).

Confirmed

This field informs whether the posting has already been uploaded to a higher-level system. Y = uploaded, N = not uploaded, X = blocked

Confirmation date, Confirmation time (only with tabular view)

Date and time when the posting has been uploaded.

Wage confirmed

This field informs whether data is uploaded to a payroll system according to the customer's requirements.

Date of authorization, Authorization time (only with tabular view)

Date and time when the posting was last authorized.

Reference

Unique reference of the posting

Type

This field informs how this data record was collected or changed.

(empty): Original posting from data collection.

E: Manually created or changed posting. Postings that have been edited (type "E") are not copied and stored as a type "O" posting.

O: If original posting is edited, the original posting is copied and stored with type "O". The original information remains available and can be displayed.

S: Cancellation for PPS

Cancellation and original postings cannot be modified or deleted.

Other fields

Posting date, posting time

Reserved for future add-ons.

Insert new posting

Function authorization	oboo.create
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If you manually create a posting of record type "U" or "E", a posting of record type "T" is automatically created. The posting of record type T includes the quantities of the U/E posting, but no times. The end time of the U/E posting is used as point in time.

Automatic status change of the operation

- If a posting of record type "E" is created for an operation, the operation is set to the operation status "E", if the operation status was "V" or "U". Additionally, the points in time of the "first logon" or the "last logoff" are set in the order status, if these times are not yet available.
- If a posting of record type "U" is created for an operation, the operation is set to the operation status "U", if this status was "V". Additionally, the point in time of the "first logon" or the "last interruption" are set in the order status, if this time is not yet available.
- Note: In both cases, the processing is not identical to the processing of an operation interruption (A_UN) or an operation logoff (A_AB/A_BE).
- The points in time that are saved for "first logon", "last interruption" or "last logoff" are not changed in the order status, if the operation has already been started. In this case, the date is the actual date of the real, first logon.

Automatic change of quantity of the operation

- If a posting of record type "E" or "U" is created for an operation, the order quantity is corrected in the order status.
At the same time, a posting of a part quantity (posting of record type "T") including the quantity is created.
- If a posting of record type "T" is created for an operation, the order quantity is changed in the relevant order posting of record type "U" or "E" and order quantities are revised in the order status.

- If a posting of record type "E" is created, target quantities are not compared.

Change posting

Function authorization	oboo.edital
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A cancellation record is generated if a posting is already uploaded to the PPS system (confirmation flag "J"). In any other case, the data record is changed physically.



Note: When you change a posting, there must not be any running ERP uploads.

Automatisms when postings are changed

- The original data record is saved if the posting is generated automatically.
- You cannot change cancellation bookings and original bookings.

Changing postings that are already uploaded

You can configure in the [Order type](#), if changes made in the editing functions are uploaded to the ERP system. The following options are available:

- J Allow modification and upload
- E Allow modification; no upload – data may be changed but not uploaded.
- N No, modifications are not allowed – data can no longer be changed.

The check is performed in the order type of the order that you want to change.

If the option is set to "J" and you change an already uploaded posting, a cancellation posting is created. This cancellation posting keeps the original reference. The changed posting is created with the new reference.

Note: No check is performed for MDE postings.

Changing the record type

The record type of a posting cannot be changed from

- record type "H" / "U" / "E" to record type "B"
- record type "B" to record type "H" / "U" / "E"

A validation check rejects the change.

Correcting the order number in a posting

The below-mentioned procedure/order has to be observed to change the order number in a posting:

- Copy the posting to be changed to the new order number.
First copy the order-related posting (record type "U", "E") and then the staff posting (record type "B").
- Delete the postings made for the wrong order number.

Changing the quantity for an order posting (record type "U"/"E")

- If the quantity is reduced, a posting of a part quantity (posting of record type "T") including the difference quantity is generated. Here, the posting includes negative quantities.
- If the quantity is increased, a posting of a part quantity (posting of record type "T") including the difference quantity is generated. In this case, this posting includes positive quantities.
- The scrap reason of the U/E-posting is used as scrap reason for the new posting (record type "T").
- Order quantities are corrected in the order status
- Target quantities are not compared if a posting of record type "E" is corrected.
- If you use the Material and Production Logistics (HYDRA-MPL), quantities may ONLY be corrected within the H records and not within the U/E postings!

Changing quantities for a posting of a part quantity (record type "T")

- The quantity is changed in the respective order-related posting (record type "U" or "E").
- The batch record (posting of record type "H") is changed if the operation requires batch management.

Changing quantities for a batch posting (record type "H")

- If the quantity is reduced, a posting of a part quantity (posting of record type "T") including the difference quantity is generated. Here, the posting includes negative quantities.
- If the quantity is increased, a posting of a part quantity (posting of record type "T") including the difference quantity is generated. In this case, this posting includes positive quantities.
- The quantities are adjusted in the U/E record.
- The batch status is not adjusted!
- If you use the Material and Production Logistics (HYDRA-MPL), quantities may ONLY be corrected within the H records!

Changing quantities for a staff posting (record type "B")

Quantity changes for staff postings generally do not affect the order status or order-related postings.

Changing the time in an order posting (record type "U"/"E")

- If times (logon/logoff) of an order posting (record type "U" or "E") are changed, the times of the posting(s) of part quantities (record type "T") are also changed, if this/these posting(s) would then no longer be within the order posting.

- If you want to shorten the times (logon/logoff) of an order posting (record type "U" or "E"), you must first adjust the times of staff postings (record type = "B") that might be included. As otherwise, there would be staff postings outside of order postings that could no longer be changed.
- The points in time of the "first logon", "last interruption" or "last logoff", which are saved in the order status, are no longer adjusted, if you change the points in time (logon/logoff).

Changing the time in a staff posting (record type "B")

- If points in time (logon/logoff) of a staff posting (record type "B") are changed so that the middle of the staff posting is outside of an order posting, an error message will occur because a staff posting must not exist without a respective order posting.

Procedure for changing times in posting records

- If an order posting is shortened, you must first shorten the staff postings that might relate to it.
- If you want to extend a staff posting (longer than the original order posting), you must first extend the related order posting.

Changing RPAs in staff postings (record type "B")

- The personal resource performance accounts are changed accordingly in the order posting. This change is always made, irrespective of the option Proportionate RPA posting in staff postings in the HYDRA basic settings.
- In the section *Personal resource performance accounts*, the labor time is changed, if an RPA is changed and if the option *Offset labor utilization* is enabled. The total of the RPAs without RPA 12 (BKS - breaks) is displayed in the field *Sum (without breaks)*. The total of the RPAs can deviate from the labor time because labor times can be changed manually.



Use the following [INI configuration](#) to permanently activate the option *Offset labor utilization* in the editing dialog:

Field	Value
Name	BDE
MOC user	0
Section	CALC_LABOR_UTILIZATION
Key	SET_CHECKBOX
Value	TRUE
Active	☒
Comment	Set checkbox to calculate labor utilization in order bookings

Labor time / personal RPAs in order-related postings (record type "U"/"E")

- In order-related postings, the field "labor time" is not automatically changed if you make changes in the staff postings. You must change the labor time manually in the order booking.
You can manually change the duration or use the option *Offset labor utilization* to integrate the changes made as described in the paragraph Changing RPAs in staff postings (record type "B").
- If a staff posting is changed, the personal resource performance accounts are changed accordingly in the order posting. This change is always made, irrespective of the option Proportionate RPA posting in staff postings in the HYDRA basic settings.

Copy posting

Function authorization	oboo.copy
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If you use the copy function, you create a new posting. The only difference to a completely new posting is that the data of the copied posting is entered into the fields of the dialog box.



With postings of record type U, E, H or T, the preassigned values of the person-related resource performance accounts are not saved.

Delete posting

Function authorization	oboo.delete
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- If a posting has already been uploaded to the PPS system (confirmation flag "J"), then a cancellation record is created. Otherwise, the data record is physically deleted.
- The operation is reactivated automatically if a posting of record type "E" is deleted.
- If you delete a posting of record type "U" or "E", all included postings of part quantities (record "T") are automatically deleted.
- Staff postings (record type "B") are not deleted when you delete order-related postings (record type "U"/"E"). Consequently, they should be deleted manually *before* deleting order-related postings.
- If the HYDRA basic setting Maintenance on the basis of events is enabled, then you can only delete manually created postings.
- The saved points in time "first logon", "last interruption" or "last logoff" are not updated in the order status if an order-related posting is deleted (record type "U"/"E").

Authorize postings

Function authorization	oboo.sign
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When you authorize a posting, the date of authorizing and the authorizing person are set.

If the data record is also included in the [Foreman's checklist](#) in the category *Non-authorized personnel postings* or *Non-authorized order postings*, this posting is identified as *Settled* in the foreman's checklist.

If an open [Escalation](#) "ANR.UNCERTIFIED_BOOKINGS" exists for the posting, the escalation is closed with the comment "booking signed".



You can define for each [Order type](#) that postings should only be uploaded after authorization. After 5 days at the latest (standard setting, [can be changed](#)), non-authorized postings are also confirmed/uploaded.

General validation checks

- If a posting has already been uploaded and the option *No change after upload* is set, you cannot change or delete the posting.
- If a posting has already been uploaded and the configuration Change allowed, no upload is enabled in the order types, then you can change or delete the posting after confirming it in an information dialog. In this case, the status is corrected in HYDRA but an upload/cancellation is not sent to the higher-level system.
- You can only edit or delete postings that have been recorded/generated automatically if the event maintenance function is disabled. But you can always change or delete posting records that have been created manually.
- You cannot change or delete cancellation postings (record type "S") and original postings (record type "O") because these postings are only created for reasons of traceability.
- If you insert/edit/copy order postings (U/E records), the system checks whether the times overlap with already existing order postings. The period, order number and machine are checked in this context.



If record type "E" or "U" is entered, the system does not check if at the time of the generation or change of the posting the operation was logged on, already finished or interrupted or not logged on at all.

- If you insert/edit/copy a posting of a part quantity (record type "T"), it is checked whether a corresponding order posting exists.
- If you insert/edit/copy a batch posting (record type "H"), it is ensured that
 - a corresponding order posting exists
 - the operation requires batch management
 - no overlapping batch posting exists for the order/workplace.
 - either yield or scrap can be entered, but not both quantities.

- The user must be authorized for the responsibility area that includes the workplace indicated in the posting record.
- When you change a posting, there must not be any running uploads.
- If you use the editing function for postings to change automatically collected data or to create completely new postings (e.g. external orders), it is checked whether or not the workplace is currently blocked by the event maintenance function.

20 Event Maintenance

Overview

	Drei kleine textuelle Korrekturen/Änderungen.	
Menu	Data collection → Data correction → Event maintenance	
Transaction code	evli	
Function authorization	evli	

Purpose

You can use the *Event maintenance* to edit and recalculate the posting events provided by the data collection. You can also delete events or create new events.

Integration

The user can use the maintenance function to subsequently change events. The system can then perform a validation check and recalculate the events. The resulting postings can be integrated (selected) in the different evaluations or uploaded to the higher-level ERP system.

The special feature of the event maintenance, compared to the editing functions for postings, is that it includes a recalculation function. The changed data (events) is then used to generate new postings. When data is recalculated, the system performs the same validation checks as for postings of the online data collection. This procedure ensures consistent data.

Requirements

The event maintenance provided with this application requires activation. If the activation is not performed, the following message is shown when the application is called: "The event-related BDE maintenance has been disabled in the basic settings. Modifications cannot be saved." Without activation, the events can only be selected and displayed, but you cannot add new events or edit or delete existing events.

By default, the event maintenance is not active.



Only activate the event maintenance if the relevant requirements or the restrictions listed in the following have been checked.

Restrictions

Some posting functions are pure data collection functions. The collected data cannot be edited or recalculated in the event maintenance. You must check for each function whether the event maintenance can be used. Find a list of these functions in the following.

Data from data collection that cannot be edited

- Comments entered in different posting dialogs on the shop floor terminal cannot be displayed or edited in the event maintenance.
- Serial numbers collected for OPs that require serial numbers cannot be displayed or edited in the event maintenance.
- The system can collect quantities in alternative units. In the event maintenance, you can only edit quantities in primary quantity unit.
- You can post manual activities in the system or calculate them using formulas. You cannot use the event maintenance to show or edit activities.
- Posting with OP reference or confirmation/upload number: In the event maintenance, you require the MES order number for editing.
- Changes of the production lock are logged as events. You can display these events in the event maintenance, but you cannot edit them.
- User fields that were integrated in posting dialogs as part of a customization (e.g. OP logoff) cannot be displayed or edited in the event maintenance.
- Automatic posting processes and their configurations are not used and recalculated in the event maintenance (e.g. finish/interrupt or automatically log off a preceding operation when target quantity is reached).
- Events that are based on resource-related postings (WRM or DNC) cannot be displayed or edited in the event maintenance.
- The "Automatic OP change" option in tab *BDE* of the *HR master data* is only integrated during online processing; it is not integrated during recalculation in the event maintenance.
- The option "OP change with status change" in the status assignment of machines/workplaces is only integrated during online processing; it is not integrated during recalculation in the event maintenance.
- Postings of personnel or machine waiting periods are not displayed in the event maintenance.
- The recalculation in the event maintenance does not trigger escalations.
- Optional data that has additionally been recorded in the "Change machine status" dialog is not displayed in the editing dialog and cannot be changed. This additional data can be the "expected duration", a comment or any additional data for escalation management.

Merged operations

An assignment of a new OP to a merged operation is NOT supported in the event maintenance. Only data for already posted merged operations can be modified.

If data is recorded for a merged operation using the "Merged operation per machine" function, this data CANNOT be edited in the event maintenance.

A recalculation is no longer possible for merged operations that were generated on the MOC if the operations contained in the MOP were changed.

PZE controls ADE

The "PZE controls ADE" setup setting can lead to scenarios in the event maintenance where the personnel logon event is displayed before the actual order logon, because the logon is then rounded to the PZE time.

If inconsistent postings exist in the event maintenance – because a personnel logon is displayed before the actual order logon – then the postings must be corrected. The personnel logon time must be changed to the time of the order logon.

Interface to SAP

The event maintenance cannot be used in combination with the PP-PDC interface, because the standard SAP system does not accept cancellations from external systems.

MDE shift change events

If these events are changed, it is possible that shifts are not calculated completely or that gaps result. For this reason, generally do not edit shift change events.

A recalculation of the entire shift is not guaranteed here because the relevant events that identify the times of start and end of shift might no longer be available at the relevant point in time.

Shift change events are events triggering status changes "M_MST"

end of shift → dialog A_AUN

beginning of shift → dialog A_AAN

Shift end events (event M_MST with dialog A_AUN) may only include the status 20000 "no shift".

Editing of finished operations

Operations that have already been finished can be edited in the event maintenance dialog. Note: The status of a finished operation does not change. The status is also not changed if an OP logoff is deleted or if the OP logoff is changed into an interruption.

If you want to reactivate an operation that has already been finished, you must always use the function "Reactivate operation".

Material and Production Logistics (MPL)

Some operations require batch management. The following restrictions apply for these operations in the event maintenance:

- Only MPL machines with batch processing can be edited. Machines of the types roll cutting, parallel output batches, throughput batch mode, etc. are not supported.
- The event maintenance supports input batch postings for an operation to a limited extent only. Postings that are not included in the OP posting are not integrated. Consequently, consumption postings for input batches are not integrated.
- It is not allowed to change output batch numbers or to delete events CA_AN/CA_AB.
- Using the event maintenance, you cannot display or change batch attributes.
- The existing batch tracing is not changed subsequently by the maintenance function.

Event modified by

You can store a user ID for the event. For technical reasons, the user ID has a maximum of 9 digits (column "Modified by"). This means: If you enter a user ID with 10 digits, the last digit is cut when the user is transferred to the "Modified by" column in the event maintenance.

Editing of the partitioning

If a machine partitioning event (M_TLG) is edited, only the shifts included in the recalculation are integrated.

If an order is produced in several shifts (automatic re-login), **all** events must be recalculated that are affected by this change of partitioning event.

Respect the following process for an event that includes a change of the partitioning:

- If partitioning is changed, the entire period for which this modification applies needs to be selected, i.e. until the operation is interrupted or finished manually.
- When changes are made, the relevant partitioning event must be changed
AND
if the operation was automatically produced during several shifts, the subsequent manual interrupt or finish posting of the order must also be changed.
If you manually interrupt or finish the order, it is enough to save the data without change.

- When data is then recalculated, all data between the change of partitioning and the manual interrupt or finish posting is automatically used for this recalculation.

Note:

The events that must be recalculated are not selected automatically, because the change of partitioning can affect several shifts; also shifts that are no longer included in the selection period.

Selection criteria

The application provides the following selection criteria:

Workplace/machine

All data relating to the selected workplace is shown.

MES order number

The result list shows all events of workplaces where the order was logged on in the selected period of time. You can also use wildcards.

Person ... to ...

The result list shows all events of the workplaces where the person (persons) was (were) logged on in the selected period of time. You cannot use wildcards.

Date ... to ...

All events that are included in the selected period of time are displayed.



Note: This application only selects and edits data that is included in the online data area.

Parallel staff logins

The option "Parallel staff logins" selects the data of all workplaces where a person was logged on. Set this option if you want to edit data of workplaces where persons are logged on with multiple machine operation.

For more information, see [here](#).

Refresh data

Use this option to control if the new data is displayed in the event maintenance after recalculation (with large amounts of data, this can have a negative effect on performance). Note: If an error occurred during recalculation, the data freezes in the display.



If more than 1000 events are selected according to the data selection, the following error message is displayed: "The requested amount of data is too large. Please restrict the data using selection criteria."

When data is requested, all selection fields are disabled except the checkbox "Refresh data".

The fields are only enabled again when the button "Discard" is clicked.

If you request data and this data is currently locked by another user, then you can specify if the data is displayed. Here, the data is only displayed and cannot be edited.

Toolbar

You can call the following functions from the toolbar. Note the [Requirements](#) for editing events:



Edit an existing event

Function authorization: evli.edit

The relevant dialog for the event selected is opened.



Delete an existing event

Function authorization: evli.delete

The selected event is deleted.



Recalculate

After the confirmation prompt, a validation check is performed for all added, edited or deleted events; the postings resulting from the events are regenerated. When the recalculation is completed, a message is issued.

If an error occurs during recalculation (e.g. because of logically invalid values), an error message is displayed.



Discard

After confirmation, all added, edited or deleted events are discarded and the list is cleared.



Order information

Function authorization: orin

Calls the application [Order information](#).



Workplaces/machines

Function authorization: wpov

Calls the application [Workplaces/machines](#).

Tab Create event

Function authorization: evli.create

This tab includes the buttons  to create events. Different categories are available:

Operation category:

- Log OP on (A_AN)
- Log OP off (A_AB)
- Partial confirmation/posting of part quantity (A_TR)
- Quantity upload (A_MR)
- Interrupt OP (A_UN)
- Finish OP (A_BE)

Person category:

- Log person on (P_AN)
- Log person off (P_AB)

Machine/workplace category:

- Change status (M_MST)
- Change partitioning (M_TLG),
- Automatic counter (M_CTR_AUTO),
- Automatic quantity (M_AUTO)

Material category:

- Log output batch on (CA_AN)
- Log output batch off (CA_AB)
- Log input batch on (CE_AN)
- Log input batch off (CE_AB)

Other category:

- Activate OP (NC_AN)
- Deactivate OP (NC_AB)

Field descriptions

Class

Internal event classification:

P = Personal data

M = Workplace data

A = Order data

C = Batch data

Event

The possible events and their colors are listed [here](#).

Dialog

This field shows the dialog that triggers the event. Some dialogs have the same name as the event (e.g. A_AN, A_UN, ...), but a different dialog can also trigger the event.

Date, time

Date and time when the event was posted.

Workplace/machine

The event was posted for the workplace specified.

MES order number

The event was posted for the specified combined order/operation number. This field is only filled in if the event has an order reference.

Person

The person who triggered the event or for whom the event was triggered. This field is only filled in if the event was triggered by a person or has a reference to a person.

Status

The workplace/ machine status that applied at the time of the event; this field contains only context-related information.

PZE

Reference to clock-in or clock-out

Primary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Secondary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Tertiary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Base quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Modified by

Last editor of the event

Date, time

Date and time when the data record was last edited.

Editable

J = Event can be edited

N = Event cannot be edited, e.g. is the master of an MOP or is locked by a change in the log record

Reference

Unique ID of the data record

Priority

Priority specification for events of the same time (1 = highest priority)

Sorting

Internal use

Status 1

Event M_MST: machine status/ interruption reason

Event M_TLG: Partitioning

Event M_SZY: Target cycle

Status 2

Internal use

Status 3

Internal use

Attribute 1

Events C_GEN, C_UMB, CE_AN, CE_AB, CA_AN, CA_AB: target location/material buffer

Attribute 2

Event P_AN: Wage/ premium indicator

Events CE_AB, C_GEN: info on batch

Otherwise: Internal use

Attribute 3

Event P_AN: Operator position

Events CA_AB, C_GEN: transport unit

Event CE_AN: BOM item

Event CE_AB: Batch status

Attribute 4

Internal use

Attribute 5

Event CE_AB: Current batch

Counter 1, Type 1, Reason 1

Internal use

Counter 2, Type 2, Reason 2

Internal use

Counter 3, Type 3, Reason 3

Internal use

Counter 4-10, Type 4-10, Reason 4-10

Internal use

Partitioning

Partitioning. Is currently only displayed for the event M_TLG.

Displayed events

The following events are displayed:

Event	Designation	Type
A_AN	Log operation on	Order-related event
A_TR	Partial confirmation / posting of part quantity	Order-related event
A_UN	Interrupt operation	Order-related event
A_AB	Log operation off	Order-related event
A_BE	Finish operation	Order-related event
A_MR	Quantity upload You can use the quantity upload to upload quantities for operations, which are not logged on at the moment. This way, you can correct a quantity of an operation without having to log the operation on and off.  If an operation is logged on, you may not use this event. Use instead the event A_TR.	Order-related event
P_AB	Log person off	Person-related event
P_AN	Log on person	Person-related event
P_VAN	Person advance logon	Person-related event

Event	Designation	Type
		Cannot be edited. The event is not used during recalculation.
M_MST	Change workplace/machine status	Machine-related event
M_AUTO	Automatic quantity posting from the terminal	Machine-related event
M_CTR_AUTO	Automatic counter posting from terminal	Machine-related event
M_TLG	Change of partitioning	Machine-related event
M_SZY	Change of target cycle	Machine-related event Cannot be edited. If the target cycle is changed, the change is not used during recalculation.
M_PSPERRE	Production lock	Machine-related event Cannot be edited. Production lock events are not used during recalculation.
CA_AB	Log output batch off (MPL)	Batch-related event
CA_AN	Log output batch on (MPL)	Batch-related event
CE_AB	Log input batch off (MPL)	Batch-related event
CE_AN	Log input batch on (MPL)	Batch-related event
NC_AB	Deactivate OP (ADE-BEA)	Other event
NC_AN	Activate OP (ADE-BEA)	Other event

The colors of the listed events are the following:

Color	Meaning
blue	Order-related events
green	Person-related events
red	Machine-related events
brown	Batch-related events

Color	Meaning
black	Locked events
purple	If the order type option "Change after upload" is set to "Allow no changes" or "Allow modification, no upload", then all the events already uploaded to the ERP system are displayed in purple.

The column "Dialog" shows the dialog that triggered the event. Some dialogs have the same name as the event (e.g. A_AN, A_UN, ...). The following other dialogs can also trigger the events mentioned above:

Dialog	Meaning
A_P_AN	Log order + person on in one (A_AN + P_AN)
P_AAB	Log all staff off (1..n P_AB)
A_AUN	OP is automatically interrupted with shift change
A_AAN	OP is automatically logged on again with shift change
SA_AN	Log on merged operation
SA_TR	Partial confirmation for merged operation
SA_AB SA_ABME	Log merged operation off
P_KOM	PZE Clock-in
P_GEH	PZE Clock-out
CA_WL	Output batch change

Display of automatically recorded counter values:

If quantities are automatically recorded for a workplace using the Windows terminal software, then these unevaluated counter quantities are collected and posted as event M_CTR_AUTO. The counter values of the counter events M_CTR_AUTO are not displayed in the columns for yield, scrap, etc., but in separate columns of the category "counter".

The primary quantities resulting from the counter values (evaluated) are logged and displayed as information in the columns for yield, scrap, etc.



The resulting (evaluated) quantities are not logged in other quantity units. This means: The base, secondary and tertiary quantities of the counter events are always 0.

No reason is added to the logging of the resulting (evaluated) quantities.

Recalculation of changed data



When you have completed the editing, click button  and confirm the prompt to start the recalculation function. The recalculation can take some time depending on the extent of changes made.

The following steps are performed during recalculation:

1. Validation check of the modified data

The system checks if all events involved are fully processed (same check as with online processing). If the validation check fails for one of the events, then the recalculation is rejected and the changes are not accepted in the system. If a validation error occurs, then an error message is displayed including information on the relevant event.

2. Canceling existing results

If all validation checks were completed successfully, then the still available, current results are canceled. If postings have already been uploaded to the ERP system (upload identifier "J"), then the relevant cancelation records are generated; this is the same processing when postings are manually changed using the editing function for postings. The quantities and times contained in the order-related posting are also canceled for the operation status.

3. Calculating new results

After a successful validation check and cancellation, the modified events are reevaluated and a new posting is generated. The quantities and durations calculated for the order-related posting are also posted to the operation status.



Note: If the configuration has been changed in the meantime, it is possible that you cannot change events any more that have already been recorded or the changed configuration can lead to other results.

Example: If two operations were logged on to a machine at the same time, and in the meantime the option *Parallel logon of OP* was deactivated, then the recalculation of one operation leads to a validation error. Recalculation is no longer possible.

If the recalculation could be completed successfully, a success message is shown.

Locking concept in the event maintenance

The selected data is locked for editing. The selected machine and the selected period of time specify the data locked.

All locks of the event maintenance are displayed in *System administration > Locked data records* and can only be deleted by a user with the relevant function authorization.

If a lock is deleted, the recalculation of the currently displayed data is no longer possible on this console. The user must discard the scenario and request data again.

Option "Parallel staff logins" in the event maintenance

If the employees log on to multiple machines, you must activate this option in the event maintenance selection area. The personnel times are then allocated according to the order postings.

All machines are then displayed where the displayed persons were logged on during the evaluation period.

If a nesting of the data is available, this nesting might be too complex and the data cannot be recalculated because the start events are no longer included in the period of time selected.

Example:

Person 1 is logged on to workplaces 100 and 200

Person 2 is logged on to workplace 200 at a different, but overlapping time

Person 3 is logged on to workplaces 200 and 300 at a different, but overlapping time

When data is recalculated, all events of an order are integrated. With nested order events, a great number of order events must be recalculated. Here, a constellation is possible that does no longer allow a recalculation.

Note: You cannot use the event maintenance with such complex nested personnel postings.

Option "Optimized parallel staff logins" in the event maintenance

If this option is selected, the following optimized processing is performed with parallel staff logins/personnel postings:

- Only the data specified via selection is displayed or requested (machine, order/OP, person)
- Only if a recalculation is performed, the machines are selected that must be recalculated because of parallel staff logins.
The system uses the changed data to identify the relevant machines.
- If an error occurs during recalculation, the additional machines are also displayed and the user can correct the data.

Waiting period processing

Waiting period processing is an optional processing that controls the system behavior when personnel postings are collected. You activate the processing in the basic settings.

If the waiting period is exceeded, a separate waiting period posting is generated (personnel posting to the defined waiting period operation). If the waiting period is not exceeded, the personnel posting is backdated (and also the OP posting, if required).

The posting times of the events remain unchanged during waiting period processing, i.e. the posting times are edited in the event maintenance.

If the events are recalculated, the changed postings are used to identify whether the times must be backdated according to the waiting period processing.

Changes after upload

As part of the customization, you specify for each order type whether changes made in the event maintenance are uploaded to the ERP system. The following options are available:

- Allow modification and upload
- Allow modification, no upload - You can change data, but the data is not uploaded
- Do not allow modification - data cannot be changed.

The validation check is performed for the workplace: If an order at a workplace cannot be edited according to the order type configuration (purple font color), then all events are locked that are older than the last upload date/time of this order. The lock is performed for the workplace, i.e. also orders that could be edited, are locked.

21 Foreman's Checklist

Overview

Menu	Information management → Messages → Foreman's checklist
Transaction code	fmchkl
Function authorization	fmchkl

Purpose

You use the foreman's checklist to monitor the operations produced in the foreman's area:

- The list shows unusual data (e.g. time or quantity deviations)
- The list shows postings that the foreman must authorize.

The information displayed in the foreman's checklist is generated and saved in a database table on a daily basis (using an application on the server that is automatically started during the night). The list shows information on the last 7 days. The application evaluates the collected information.

Integration

The data displayed here (e.g. time, quantity deviations) is based on the data collection performed via collection units (e.g. terminals).

Selection criteria

The application provides the following selection criteria:

Order

Selection using the order number. You can use wildcards.

Category

Selection using order categories.

Order type

Selection using order types.

Workplace

The application displays the postings that are made for the workplace number entered.

Responsibility area

The application displays the postings that are made for the workplaces that are assigned to the responsibility area entered.

Cost center

The application displays the postings that are made for the workplaces that are assigned to the cost center entered.

Company

The application displays the postings that are made for the workplaces that are assigned to the company entered.

Posting

The following information is logged. If you specify the field *Posting*, you can narrow down the display to the required posting categories:

(2) Postings with target-actual quantity deviation exceeding +/- 5%

Finished operations, for which a yield (in primary quantity unit) is posted of more than +/- 5% compared to the target quantity (in primary quantity unit).

(B) Postings with target-actual time deviation exceeding 10%

Finished operations with an order duration (times that are posted to 11) of more than 10% of the target processing time. Only operations with a target processing time greater than 0 are used here.

(5) Non-authorized personnel postings

Non-authorized BDE log records for staff (record type "B"). The log records can be authorized using the application [Order-related postings](#).

Via customization, you can specify for each order type whether these postings are logged.

(6) Non-authorized order postings

Non-authorized order-related BDE log records (record tpe "U", "E") for overhead cost operations. These log records can also be authorized using the application [Order-related postings](#).

Via customization, you can specify for each order type whether these postings are logged.

(9) Open operations of finished production orders

Operations of production orders (category "FA") that are still active (prepared, running or interrupted) and with a last operation already showing a status "finished".

(1) Postings showing scrap without reason Log-offs or interruptions of operations are displayed where the (only) scarp reason is 999. Condition: Scrap reason 999 and manual scrap postings on the terminal with scrap reason 999 must be configured.

Show finished messages

If this checkbox is not set, only pending postings are shown. If the checkbox is enabled, completed and pending postings are shown.

In general, only data is output if the user is authorized for the relevant responsibility area. If you select operation data, the responsibility area of the machine/workplace is used where the order was produced. If you show staff data, the responsibility area of the person is used as selection criterion.

Field descriptions

The foreman's checklist shows the postings generated for a day in a table. The table shows information on the posting and the connected master data (workplace, order, persons) that is read from the tables stored.

The totals line shows the number of entries in column "Posting".

Posting category

Posting: see paragraph [Selection criteria](#).

Production date/production time: the data's origin for this value depends on the type of posting:

- Postings of finished operations with target/actual deviation: time of most recent log-off
- Open operations of finished production orders: time of the last posting (for running and interrupted operations), otherwise empty
- Non-authorized personnel/order postings: Log-off time of the log record
- Postings showing scrap without reason: Log-off time of the log record

Operation category

Order

Operation

Article (of the operation)

Article designation

OP name

Other category

Workplace

Short name (of the workplace)

Person (personnel number)

Name (first and last name)

Last name

First name



Personnel data is shown with the following postings:

- Non-authorized personnel postings
- Non-authorized order postings if a staff badge number has been entered with logoff/interruption.
- Postings of scrap without reason if a staff badge number has been entered with logoff/interruption.

The other postings do not provide any personnel data; the (cumulated) status information of the operation is used here.

Order category

Evaluation date (start of the evaluation period; the start of the evaluation period is calculated as follows: "today" - INTERVAL. If you call the hy_mst application on the server, INTERVAL can be transferred as a parameter. If this parameter is not transferred, 7 is assumed as the default value).

Category (order category)

Order type

Final article (article of the order header)

MRP controller

Order group

Project number

Sequence

Split

First logon (of operation, date/time)

Last posting (for the operation, date/time)

Target quantity: target quantity of the operation (primary quantity unit), only filled if a "target/actual quantity deviation" is posted.

Actual quantity: yield of the operation (primary quantity unit), only filled if a "target/actual quantity deviation" is posted.

Target duration: target processing time of the operation, only filled if a "target/actual time deviation" is posted.

Actual duration: main utilization time posted onto the operation (RPA 11), only filled if a "target/actual time deviation" is posted.

Workplace category

Designation

Group

Cost center

Company

Responsibility area

Person category

Company

Cost center

Area

Department

Responsibility area

Signing category

Done

Shows whether or not the entry has already been marked "done" by a user.

Modified by

ID of the user who marked the entry as "done".

Done on

Point in time when the user marked the entry as "done".

Toolbar**Go to category****Order information**

Function authorization: orin

Calls the [Order information](#). The order number is transferred as parameter.

**Order overview**

Function authorization: orov

Calls the [Order overview](#). The order number is transferred as parameter.

**Workplaces/ machines**

Function authorization: wpov

Calls the application [Workplaces/machines](#). The workplace number is transferred as parameter.



Order related postings

Function authorization: oboo

Calls the application [Order-related postings](#). The following values are transferred as parameters:

Workplace, order, operation, production date.



Done

Function authorization: fmchkl.sign

Selected postings can be marked "done" by this function (multiple selection is possible).

Processing notes

The information listed in the foreman's checklist is generated and saved in a database table on a daily basis (by an application started on the server automatically overnight). This database table is then used by the MOC application. The checklist includes information on the last seven days (default setting).

If required, the below-mentioned call parameters can be added to the application hy_mst.exe/.out that is integrated in the HYDRA Scheduler:

/INTERVAL=days

Increase evaluation period (default: "today" - 7 days)

/DEL_SIGNED= days

Delete authorized postings after <days> (default: 7 days)

/DEL_UNSIGNED= days

Delete non-authorized postings after <days> (default: 999 days)

/NO_CERT

Non-authorized personnel postings are not integrated in the foreman's checklist.

For this parameter, you can specify via customization which order types

- personnel postings
- order postings

must be authorized and are then shown in the foreman's checklist.